An IoT-Enabled System for Real-Time Fuel Monitoring, GPS Tracking, and Anomaly Detection in Trucks Using Non-Parametric Unsupervised Machine Learning

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ABSTRACT

The logistics and transportation industry continues to face challenges in fuel management, anomaly detection, and real-time fleet monitoring. Existing solutions often operate in silos, resulting in fragmented data, undetected fuel irregularities, and delayed maintenance responses. This study developed and evaluated an IoT-based, context-aware fleet monitoring system that acquires real-time fuel level via OBD-II, tracks GPS location, and detects anomalous fuel consumption in trucks using non-parametric unsupervised machine learning models, including Matrix Profile and Isolation Forest. The system leverages CAN Bus/OBD-II interfacing for ECU fuel data, GPS modules for location logging, LoRa and GSM modules for reliable IoT-based communication, and embedded alert hardware for driver rest and maintenance notifications. Sensor and vehicle data are synchronized and transmitted to a cloud-based dashboard where anomalies are automatically flagged, and actionable alerts are generated. The system will be evaluated in real-world driving scenarios to measure detection accuracy, false positive rates, and response latency. This integrated platform aims to enhance operational safety, improve fuel efficiency, and enable proactive, data-driven fleet management decisions. Three user-facing interfaces were delivered: an embedded in-truck HMI for drivers, a mobile driver web application for companion GPS and trip-state synchronisation, and an administrative dashboard for SGSA fleet monitoring.

Index Terms—Fuel Monitoring, GPS Tracking, Anomaly Detection, Internet Of Things, Unsupervised Model Learning, Isolation Forest, Matrix Profile

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ABBREVIATIONS

GSM	Global Systems For Mobile Communications	5
IOT	Internet Of Things	5
RFID	Radio Frequency Identification	4

NOTATION

L	data transmission latency	121
TP	true positive	121
FP	false positive	121
TN	true negative	121
FN	false negative	121
T_{detected}	timestamp when an anomaly is detected	121
T_{actual}	timestamp of actual anomaly occurrence	121

233 # GLOSSARY

234	cloud analytics	Data analysis performed on cloud-based platforms, enabling real-time access to fleet data (e.g., GPS and fuel metrics) and supporting informed decision-making.
235	deep learning	A subset of machine learning involving deep neural networks with multiple layers. Used in predictive analytics and anomaly detection in real-time fleet systems.
236	fleet management	The administration and coordination of commercial vehicle operations, including monitoring of fuel use, driver behavior, vehicle maintenance, and routing.
237	Isolation Forest	An unsupervised machine learning algorithm for anomaly detection that isolates outliers based on random partitioning of data points.
238	Matrix Profile	A time-series data mining method used for pattern recognition and anomaly detection in sequential datasets like fuel usage over time.
239	unsupervised machine learning	A type of machine learning where the model learns patterns from unlabeled data. Useful for anomaly detection where normal vs. abnormal behavior is not predefined.

Chapter 1

INTRODUCTION

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1.1 Background of the Study

The transportation and logistics industry is a critical driver of economic development but is increasingly facing operational inefficiencies and security challenges, especially in fuel management and vehicle tracking. Fuel theft, inaccurate fuel monitoring, inefficient routing, and lack of real-time data remain persistent issues for fleet operators, particularly in developing countries. These challenges are exacerbated by the absence of scalable, integrated technological systems that provide reliable data-driven solutions [Ahmed et al., 2017, Alquhali et al., 2019, Crisgar et al., 2021].

Fuel is one of the most substantial and recurring expenses in logistics operations, with its accurate monitoring being essential to maintaining profitability and operational efficiency. Even minor inconsistencies in fuel tracking—such as errors in manual readings, unreported consumption, or unnoticed leakage—can accumulate into significant losses over time. Komal Shoukat Ali et al. (2018) [Komal Shoukat Ali et al., 2018] report that fuel-related fraud and inefficiencies can contribute up to 30% of the total operational overhead in large fleet operations. These losses are often compounded by issues such as unauthorized vehicle use, fuel siphoning, and unoptimized driving routes, which not only drive up fuel costs but also disrupt delivery schedules, vehicle maintenance cycles, and driver accountability. Despite efforts to manage these concerns, many logistics providers continue to rely on fragmented or outdated systems that offer limited visibility into fuel-related metrics and driver behavior [Dhivyasri and Mariappan, 2015, Obikoya, 2014].

These pressures are especially acute in the Philippine setting, where fleet operations contend with some of the most severe road and logistics conditions in the region. The Japan International Cooperation Agency estimates that traffic congestion costs Metro Manila roughly PHP 3.5 billion per day, a figure projected to reach PHP 5.4 billion per day by 2035 without intervention [Japan International Cooperation Agency, 2022, Japan International Cooperation Agency, 2025]. Logistics operations are further constrained by congestion at the Port of Manila, which has operated at roughly 94% utilization against an ideal of about 80%, and by regulatory measures such as the 2014 Manila truck ban, which the Philippine Institute for Development Studies estimated caused around PHP 43.85 billion in economic losses and doubled container trucking costs [Patalinghug et al., 2015b, Llanto, 2017, Patalinghug et al., 2015a]. Compounding these conditions, fuel fraud and adulteration in the national supply chain have been estimated to cost up to USD 750 million annually [Asian Development Bank, 2016]. For trucks operating under such conditions, prolonged idling, stop-and-go consumption, and route variability are routine rather than exceptional, which makes fixed-threshold fuel-monitoring rules, calibrated for free-flowing conditions, especially unreliable.

In addition to financial implications, traditional fuel monitoring systems struggle to detect abnormal fuel consumption patterns in real time. These anomalies may be caused by aggressive or inefficient driving habits, extended engine idling, mechanical malfunctions (such as injector faults or fuel line leaks), or variable environmental conditions like terrain and traffic. However, without intelligent tools that correlate fuel data with contextual factors—such as vehicle speed, route topography, and time of day—fleet managers are often unable to distinguish between acceptable fuel use and problematic deviations. As highlighted by Mumcuoglu et al. (2023) [Mumcuoglu et al., 2023], conventional systems lack the sophistication to perform this level of analysis, leading to delayed detection of inefficiencies and reduced responsiveness. This underscores the urgent need for smart, data-driven solutions that combine fuel monitoring with GPS tracking and anomaly detection models to proactively manage consumption and enhance overall fleet performance [Barbado and Corcho, 2022, Beteto et al., 2022, Fortela et al., 2023].

To address these problems, researchers and engineers are increasingly exploring the convergence of IoT, GPS, and AI—particularly machine learning—to develop smart, real-time monitoring systems [Alquhali et al., 2019, Joshi et al., 2024, Prabhu et al., 2022]. These systems aim to optimize fuel usage, enhance fleet security, and improve operational efficiency through automation and predictive analytics.

The transportation and logistics sector has attracted significant scholarly attention, particularly in the development of smart systems for operational efficiency and security. The academic literature reflects a growing interest in leveraging IoT, GPS, and AI technologies to enhance fleet management, particularly in areas such as fuel monitoring, vehicle tracking, and anomaly detection. Several studies in recent years have investigated IoT-enabled fuel monitoring, GPS tracking, and anomaly detection solutions [G et al., 2022, G, 2023, Patil and Patil, 2024, Prabhu et al., 2022, RS et al., 2024, S et al., 2024].

Because partner HINO in the Philippines do not permit tank-cap or sender-loom modifications on production trucks, any field-deployable monitoring approach must be non-invasive. This constraint was the dominant driver of the acquisition-layer design adopted in this study.

- **Intelligent IoT-Enabled Real-Time Monitoring System for Logistics Management:** Joshi et al. [Joshi et al., 2024] developed a hybrid logistics management system that integrates IoT sensors, GPS, Radio Frequency Identification (RFID), and cloud analytics. Their solution achieved real-time tracking of assets and vehicles, enhancing transparency and decision-making in logistics operations. However, the study was limited by the type of hardware used and did not include advanced AI components for anomaly detection.

- **Fuel Consumption Classification for Heavy-Duty Vehicles:** Mumcuoglu et al. [Mumcuoglu et al., 2023] employed machine learning to classify fuel consumption anomalies based on naturalistic driving data from 57 trucks. With an accuracy of 92.2%, their approach effectively identified abnormal fuel consumption linked to driver behavior. Nevertheless, the dataset was relatively small (606 records), and integration with real-time vehicle diagnostics was not tested.
- **Development of an Intelligent Fuel Monitoring and Theft Detection System Equipped with GPS & GSM Integration:** Mathi et al. [Mathi et al., 2024] introduced an Internet of Things (IoT)-based solution combining resistance-based fuel level sensors, GPS, Global Systems for Mobile Communications (GSM), and real-time alarm mechanisms. Their system successfully monitored fuel and battery anomalies in hybrid vehicles but lacked higher-precision sensors and deep learning integration for smarter detection.

Despite these advancements, several limitations and research gaps remain:

- **Scalability Constraints:** Many existing solutions are limited to small-scale deployments or specific vehicle types (e.g., motorcycles, generators) and have not been validated for large truck fleets in diverse environments [RS et al., 2024, Tatababu et al., 2024].
- **Data and Sensor Limitations:** Several studies rely on basic sensors (resistance, ultrasonic) that may lack the accuracy and robustness needed for high-volume com-

mercial operations. Furthermore, datasets used are often small or simulated, limiting generalizability [Nada, 2017, aut, 2019].

- **Fragmented Solutions:** Many systems only address a single issue — fuel monitoring, GPS tracking, or theft detection — without offering a comprehensive, unified solution integrating these components using AI and IoT [G et al., 2022, Crisgar et al., 2021].

These gaps underline the need for a robust, scalable, and integrated IoT-enabled system capable of monitoring fuel levels in real time, tracking vehicle location, and detecting operational anomalies using machine learning algorithms. Such a system holds the potential to significantly reduce fuel fraud, enhance operational efficiency, and support more informed decision-making in fleet management. This study focuses on designing and implementing such a system within an active fleet operation, specifically addressing the technical and logistical challenges encountered by commercial truck fleets operating across varied conditions. By leveraging IoT sensors, GPS modules, and unsupervised machine learning-based analytics, the proposed solution aims to deliver a comprehensive, automated, and intelligent monitoring platform that advances the current landscape of smart logistics technologies.

1.2 Prior Studies

TABLE 1.1 LIST OF PRIOR STUDIES

Literature Title (APA Citation)	Research Problem	Method	Results	Limitations
Intelligent IoT-Enabled Real-Time Monitoring System for Logistics Management [Joshi et al., 2024]	Logistics systems lack integration and real-time monitoring capabilities for secure and efficient operations.	Hybrid IoT-based logistics management combining sensors, GPS, RFID, and cloud analytics for real-time monitoring and decision-making.	Provides real-time tracking of assets, vehicles, and equipment.	Constrained by hardware type and limited test cases; lacks advanced AI for anomaly detection.
Real Time Automatic Vehicle Monitoring System Using IoT [G et al., 2022]	Lack of real-time vehicle tracking and passenger/driver identity verification in institutional transport.	Used RFID, GPS, and NodeMCU ESP8266 with cloud storage and mobile app.	Enabled real-time tracking of location, speed, and passenger count.	Limited to institutional use; lacks anomaly detection and commercial scalability.
GPS-Based Vehicle Tracking and Theft Detection Systems using Google Cloud IoT Core & Firebase [Crisgar et al., 2021]	Ineffective vehicle recovery due to delayed or limited GPS/GSM tracking.	Implemented GPS-based tracking using Google Cloud IoT Core, MQTT, Firebase, and PWA interface.	Enabled real-time vehicle tracking, theft detection, and remote monitoring.	Designed for passenger cars; may require hardware adaptation for trucks/fleets.
Fuel Quantity and Quality Detection in Automobiles Using IoT [RS et al., 2024]	Fuel fraud through inaccurate dispensing and poor fuel quality in motorcycles.	IoT-enabled fuel management using strain gauges and quality sensors with real-time display/logging.	Achieved accurate real-time quantity and quality measurement.	Designed for motorcycles; not validated for larger fleets.
Fuel Consumption Classification for Heavy-Duty Vehicles [Mumcuoglu et al., 2023]	Identifying abnormal fuel consumption due to driver behavior or system faults.	Machine learning models using driving data from 57 trucks, considering weight and road slope.	92.2% accuracy; F1 score 0.78 for outlier detection.	Dataset limited to 606 records; lacks real-time integration.
Automatic Fuel Tank Monitoring, Tracking & Theft Detection System [Komal Shoukat Ali et al., 2018]	Inefficiency of manual fuel monitoring.	Fuel sensors, Arduino, GSM, GPS, LabView for data logging.	Real-time monitoring, alerts for theft/leakage.	Tested on basic hardware; lacks advanced analytics and large-scale validation.
Development of an Intelligent Fuel Monitoring and Theft Detection System [Mathi et al., 2024]	Fuel theft and battery anomalies in hybrid/electric vehicles.	Resistance-based fuel sensor, voltage sensor, GPS, GSM, buzzer, LCD.	Reliable monitoring and anomaly detection.	Lacks advanced sensors for enhanced leak detection.
Unsupervised Machine Learning to Detect Impending Anomalies in Testing of Fuel Economy and Emissions [Fortela et al., 2023]	Impending anomalies in fuel economy and emissions testing of light-duty vehicles.	Unsupervised ML methods (PCA, K-Means, SOM) on 34,602 test samples.	Trend detection in large datasets.	Limited to historical datasets; not validated on real-time operational data.
Smart Fuel Monitoring System for Diesel Generator [Tatababu et al., 2024]	Fuel quantity indication and theft in generators.	Ultrasonic/capacitive sensors, flow sensors, GSM, GPS, intelligent theft detection.	Accurate real-time monitoring, theft prevention.	Findings specific to tested generators; limited generalizability.

Literature Title (APA Citation)	Research Problem	Method	Results	Limitations
Design and Implementation for Trucks Tracking System Using GPS Based on Semantic Web [Nada, 2017]	Inefficient GPS-based truck tracking for fleet management.	GPS tracking system integrated with semantic web platform.	Accurate location tracking and reporting via web interface.	Dependent on GSM/GPS reliability; lacks IoT integration and anomaly detection.
Automatic Fuel Monitoring System [aut, 2019]	Accurate fuel monitoring and theft prevention.	Fuel sensors and GSM module with SMS alerts.	Monitors fuel, triggers alarms, and sends GPS alerts.	Basic sensors; limited scalability and analytics in complex environments.
Interpretable Machine Learning Models for Vehicle Fuel Consumption Anomalies [Barbado and Corcho, 2022]	Detecting/explaining fuel anomalies in fleets to optimize usage.	Unsupervised anomaly detection combined with interpretable ML.	Up to 80% anomalous instances explained; potential fuel reduction up to 88%.	Only analyzed individual feature relevance; not fully integrated in real-time systems.

1.2.1 Research Gaps Identified

A review of existing literature in fuel monitoring and anomaly detection for logistics and fleet management reveals persistent gaps that point to a need for more robust, integrated solutions. Although many systems incorporate IoT elements—such as sensors, GSM, and GPS—these are often limited in scope, typically designed for specific applications like diesel generators or motorcycles, rather than being adaptable for broader fleet use.

For example, systems like the Smart Fuel Monitoring System for Diesel Generators [Tatababu et al., 2024] and the Automatic Fuel Monitoring System [aut, 2019] demonstrate basic real-time monitoring and theft detection capabilities. However, these systems heavily rely on elementary sensor technology and offer minimal integration with advanced analytics or anomaly detection methods. Similarly, the machine learning-based approach in [Mumcuoglu et al., 2023] was limited in its use of data and did not incorporate real-time or unsupervised learning techniques.

While [Barbado and Corcho, 2022] explored the potential of interpretable machine learning and unsupervised anomaly detection, their models were evaluated on constrained datasets and not integrated into real-time operational systems. [Fortela et al., 2023] also demonstrated effective anomaly detection using unsupervised learning in emissions and fuel economy testing, but again, their data was static and gathered in controlled environments. These findings were not validated using on-vehicle, operational data, which limits their applicability in dynamic, real-world conditions.

Additionally, studies such as those by [Joshi et al., 2024] and [Nada, 2017] underscore the need for real-time GPS tracking in fleet operations, but they stop short of presenting a cohesive system that incorporates GPS, fuel monitoring, and intelligent anomaly detection into one unified platform.

The core research gap, therefore, lies in the absence of a fully integrated, real-time, IoT-based system that combines fuel monitoring, GPS tracking, and anomaly detection powered by unsupervised machine learning. Addressing this gap is vital to improving operational efficiency, fuel security, and predictive maintenance in modern fleet management.

1.2.2 to Address the Gaps

This study developed and evaluated an integrated IoT-based fleet monitoring system that unifies fuel level monitoring, GPS-based vehicle tracking, and unsupervised machine learning for anomaly detection. Unlike previous solutions that focus on isolated functions or static datasets, the developed system operated in real time using vehicle telemetry acquired through the OBD-II interface and GPS module. The prototype was evaluated using controlled-environment trip data and live telemetry data to assess fuel monitoring performance, anomaly detection capability, and communication latency.

The system architecture will incorporate scalable and cost-efficient components such as

, GPS modules, and cloud-based transmission via mobile networks, ensuring compatibility with a variety of fleet vehicles and usage environments.

For anomaly detection, the system will employ unsupervised, non-parametric machine learning algorithms like Matrix Profile and Isolation Forest. These models excel in identifying outliers within time-series data without the need for labeled datasets—making them ideal for fleet scenarios where data behavior varies with routes, driver habits, or environmental conditions. Unlike traditional parametric approaches, non-parametric models are more flexible and adaptive, as they do not assume any fixed data distribution.

In addition, a context-aware dashboard will be developed to provide fleet managers with meaningful insights by correlating fuel usage patterns with routes and travel times. The dashboard will support real-time notifications such as driver rest alerts, maintenance prompts, and efficiency summaries—enabling data-informed decision-making and operational improvements.

By addressing the limitations of prior research—especially the lack of integration, real-time analysis, and adaptive anomaly detection—this study offers a practical and scalable solution for enhancing fuel management and fleet operations in dynamic environments.

1.3 Problem Statement

Efficient fuel monitoring and route tracking are essential components of sustainable and cost-effective truck fleet operations. However, existing systems face significant challenges, including inaccurate fuel measurement, disconnected data sources, delayed anomaly detection, and limited support for safety-based decision-making, as outlined below.

1. PS1: The Ideal Scenario

- Trucking companies should have real-time access to accurate fuel usage and GPS tracking data to ensure efficient operations, reduce fuel theft, and improve route planning.
- Drivers should receive timely rest alerts and vehicle maintenance reminders based on actual driving and fuel metrics to promote safety, avoid overfatigue, and extend vehicle life.
- Fleet managers should be able to detect unusual fuel consumption patterns automatically, enabling early intervention and optimized scheduling.
- All of these should operate through automated, cloud-based systems that integrate IoT sensors, machine learning, and remote dashboard monitoring.

2. PS2: The Reality of the Situation

    - Most trucking operations lack real-time visibility into fuel usage, relying instead on manual readings or delayed reports that are prone to error and manipulation.
    - GPS tracking is often siloed from other performance indicators like fuel consumption and driving behavior, making behavior-based analysis or route optimization difficult.
    - Anomalous fuel consumption events (e.g., theft, leakage, or inefficiency) are rarely detected until long after they've occurred, resulting in financial loss and operational delays.
    - Rest and maintenance schedules are typically based on fixed intervals rather than real-time usage data, increasing the risk of driver fatigue, breakdowns, and missed service opportunities.

- Existing technologies are either too costly, fragmented, or lack contextual intelligence to automatically correlate GPS, fuel, and operational metrics.

3. PS3: The Consequences for the Audience

- Truck fleet operators face increasing operational costs due to undetected fuel inefficiencies and theft, impacting business sustainability.
- Driver safety is compromised when rest is not scheduled based on actual driving patterns or system recommendations.
- Vehicle lifespan shortens and maintenance costs rise when alerts are not tied to real-time usage, resulting in unexpected breakdowns and downtime.
- Decision-makers are unable to make informed, data-driven decisions due to fragmented systems and the absence of an integrated monitoring and alert platform.
- These issues collectively hinder operational efficiency, safety, and environmental responsibility, particularly for logistics providers in developing regions lacking advanced fleet management tools.

1.4 Objectives and Deliverables

1.4.1 General Objective (GO)

GO: To develop and evaluate an IoT-based system for trucks, in collaboration with the partner truck company Hino Parañaque Subic GS Auto Inc. (SGSA), that monitors real-time fuel levels, tracks routes using GPS, and detects unusual fuel consumption using machine learning, with the goal of improving driver safety, supporting rest scheduling, and facilitating proactive vehicle maintenance through automated alerts using fuel and distance metrics, based on established service guidelines.;

1.4.2 Specific Objectives (SOs)

- SO1: To measure fuel levels using digital float sensors integrated via the truck's ECU/diagnostic connector and GPS module, ensuring non-invasive installation and real-time accuracy within ±5%.
- SO2: To transmit real-time fuel and GPS data through IoT-based communication to a cloud server, accessible through a centralized dashboard.
- SO3: To capture and log GPS location data every 5 seconds via an embedded module, correlating route information with fuel and time data for behavior tracking.
- SO4: To design and evaluate a context-aware fuel usage deviation modeling approach using non-parametric unsupervised machine learning models, like Matrix Profile or Isolation Forest, achieving 90% precision and 10% false positives in identifying anomalous fuel consumption patterns within controlled environments.
- SO5: To develop and integrate hardware-based rest and maintenance alert mechanisms (e.g., Start/Stop button, buzzer indicators) tied to the web dashboard using cumulative fuel and distance data.

1.4.3 Expected Deliverables

1.5 Expected Deliverables per Objective

TABLE 1.2 EXPECTED DELIVERABLES PER OBJECTIVE

Objectives	Expected Deliverables
GO: To develop and evaluate an IoT-based system for trucks in collaboration with Hino Parañaque Subic GS Auto Inc. (SGSA) that monitors real-time fuel levels, tracks routes using GPS, and detects unusual fuel consumption using machine learning, improving driver safety, supporting rest scheduling, and facilitating proactive vehicle maintenance through automated alerts.	1. **Trained Unsupervised Machine Learning Model** • A validated ML model (Matrix Profile or Isolation Forest) trained on SGSA-reflective datasets for detecting anomalous fuel consumption. 2. **Web-Based Monitoring Interface** • Real-time dashboard showing GPS location, fuel level, distance, and operating hours for each truck. 3. **Embedded In-Truck HMI** • Driver-friendly interface displaying driver credentials (name, PIN, and assigned truck), current trip activity, and alert notifications. 4. **Cloud Backend Infrastructure** • Cloud-based data aggregation, processing, and automated alert generation accessible from SGSA's base. 5. **Performance Documentation** • Complete validation results under laboratory and SGSA field conditions.
SO1: To measure fuel levels using digital float sensors integrated via the truck's ECU/diagnostic connector and GPS module, ensuring non-invasive installation and real-time accuracy within ±5%.	1. Calibrated fuel-level acquisition module interfaced through the truck ECU/OBD2 connector. 2. Accuracy validation report comparing gotten data with fuel gauge levels. 3. Embedded firmware for fuel data acquisition, filtering, and preprocessing. 4. Drift, noise, and variance analysis across different tank levels and conditions.

Objectives	Expected Deliverables
SO2: To transmit real-time fuel and GPS data through IoT-based communication to a cloud server, accessible through a centralized dashboard.	1. Integrated IoT communication module with Lora as the primary communication path, Cellular Modules (LTE & GSM) as backup, with an option to connect to other existing hotspots are WIFI Connections. 2. Cloud backend with secure API endpoints for receiving fuel and GPS data. 3. Latency benchmarking report under variable network environments. 4. Optimized MQTT/HTTP transmission logic with buffering and fail-safe retries. 5. Security documentation covering packet integrity and encryption features.
SO3: To capture and log GPS location data every 5 seconds via an embedded module, correlating route information with fuel and time data for behavior tracking.	1. Integrated GPS module capturing coordinates and timestamps at 5-second intervals. 2. Data fusion algorithm linking GPS location, fuel levels, and time progression. 3. Route visualization through a map interface on the dashboard. 4. Recommended route engine with truck-ban zone awareness. 5. Database structure storing synchronized fuel–location–time datasets. 6. Behavioral analytics showing fuel usage per route segment.
SO4: Design and evaluate a context-aware fuel usage deviation modeling approach using non-parametric unsupervised ML models (Matrix Profile or Isolation Forest), achieving 90% precision and 10% false positives.	1. Labeled normal and anomalous fuel consumption datasets. 2. Implementation of Isolation Forest and/or Matrix Profile in Python/MATLAB. 3. Evaluation metrics report including precision, recall, and FPR 10%. 4. Exportable model (PMML/ONNX) for real-time pipeline integration. 5. Context-aware analytics combining slope, route, load, and time-of-day factors.
SO5: To develop and integrate hardware-based rest and maintenance alert mechanisms (e.g., Start/Stop button, buzzer indicators) tied to the web dashboard using cumulative fuel and distance data.	1. Cloud dashboard showing fuel usage, distance, operating hours, and maintenance status. 2. Rule-based alert engine for configurable driver safety and vehicle maintenance notifications, with example thresholds such as 300 km or 6-hour rest reminders and 5,000 km oil maintenance alerts. 3. Physical embedded module: Start/Stop button, buzzer/light indicators for rest and maintenance alerts. 4. Backend logic computing distance-based and time-based safety thresholds. 5. Dashboard filters for alert logs and historical events. 6. Extended safety monitoring: speed tracking and overspeed alerts.

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1.6 Significance of the Study

1.6.1 Technical Benefit

This research will introduce a context-aware, cloud-connected fleet monitoring system that integrates digital float sensors, GPS route tracking, and unsupervised machine learning to enhance real-time monitoring of fuel usage and detection of anomalous consumption events. Fuel levels will be measured using digital float sensors interfaced through the truck's ECU or diagnostic connector, with the system targeting an accuracy of ±5% compared to manual readings under controlled test conditions. Real-time fuel and GPS data will be transmitted to a cloud server with a maximum latency of three seconds using IoT-based communication modules, ensuring continuous data availability to fleet managers.

GPS modules embedded in the vehicles will log location data every 5 seconds, enabling the system to correlate fuel consumption with time, distance, and route conditions for behavioral tracking and operational analysis. The study will implement non-parametric unsupervised learning methods—specifically Matrix Profile and Isolation Forest—to identify deviations in fuel usage without the need for labeled datasets, allowing scalable and adaptive anomaly detection. In addition, a cloud-based dashboard will integrate hardware-driven rest alerts and maintenance reminders, leveraging cumulative fuel and distance data to generate intelligent break schedules and service interval notifications. Through these combined capabilities, the system aims to improve technical performance by enhancing measurement accuracy, communication responsiveness, anomaly detection reliability, and overall automation in fleet fuel management and driver safety support.

1.6.2 Social Impact

Through an implementation of automatic rest alerts and preemptive maintenance cues, the study encourages safety and health of the drivers to prevent accidents commonly caused by fatigue as well as breakdowns. The system will feature the ability to inform drivers when they are near critical or recommended limits (e.g. 300 km of continuous driving or 6 hours of operation) this will encourage healthy driving practices and compliance with regulations, especially for long distance hauls. These practices will empower the fleet managers who can systematically enforce them, relaying the culture of safety and responsibility into the industry.

In addition, the GPS-based route monitoring and fuel usage logging will help reinforce accountability and transparency, two essentials in preventing fuel misuse. This will help to build trust between management and drivers, and also allow the transport logistics to be data led in the decision making.

1.6.3 Environmental Welfare

The system will improve operational efficiency through minimizing waste in fuel, optimizing route performance and decreasing unplanned maintenance downtime. Through precise anomaly detection and maintenance forecasting, the platform enables cost effective operation of the fleet and longer life of vehicles. It will also help achieve environmental sustainability by minimizing unneeded fuel consumption and emissions from inefficient routes, fuel leaks or late maintenance. By promoting timely service interventions (e.g. oil changes at every 5,000 kilometers) the study aims to support cleaner engine performance and reduced environmental impact over time.

1.7 Assumptions, Scope, and Delimitations

1.7.1 Assumptions

1. Technical Infrastructure

    - Consistent internet connection is required to maintain real time data exchange between the IoT sensors, GPS module, and cloud sensor.
    - The system integration between the hardware and cloud based infrastructure operates smoothly without significant technical issues.
    - The system will have access to a stable power supply for an uninterrupted operation of the on board IoT system.
    - Mobile network coverage is able to handle a low latency data communication.

2. Data Accuracy

    - Digitally calibrated float sensors integrated through the truck's OBD2 are able to provide fuel-level measurements with a validated accuracy of ±5% when compared to manual dipstick readings.
    - The GPS module captures and logs precise coordinates at 5-second intervals, enabling accurate route reconstruction and correlation with fuel consumption and time data.

3. Machine Learning Model Performance

    - The unsupervised learning models (e.g., Matrix Profile, Isolation Forest) are capable of identifying anomalous fuel consumption with 90% precision and 10% false positives in controlled environments.

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- The training and evaluation data of anomaly detection will be representative of normal fuel consumption and truck behavior under different operating conditions.

4. User Interaction

- It is assumed that users (e.g., truck drivers, fleet operators, and system administrators) will interact with the system according to their roles and available tools.
- Truck drivers are not required to use personal mobile devices; instead, they will rely on a simple onboard interface consisting of a small display and buzzer for receiving rest alerts, maintenance reminders, and status notifications. However, a mobile web application was also developed as a supplementary HMI component to support navigation and provide additional convenience for drivers with personal mobile devices.
- Fleet operators and system administrators will interact with the cloud-based dashboard using standard office computers to monitor fuel data, GPS routes, alerts, and maintenance logs.
- Fleet personnel are expected to follow system recommendations (e.g., rest intervals, maintenance reminders) based on real-time fuel, distance, and performance metrics.

5. Operating Environment

- Environmental conditions, together with power and network breakdowns, will not significantly affect system performance.

- System maintenance is performed routinely and at normal intervals to ensure operational continuity.

1.7.2 Scope

The scope of the study encompasses the development and implementation of a smart fleet monitoring system that utilizes IoT and machine learning technologies for enhanced fuel usage , route monitoring and vehicle maintenance reminder.

1. **Real-Time Fuel Monitoring**

   a) Deployment of a calibrated fuel-level acquisition module interfaced through the truck's ECU/OBD-II diagnostic connector, providing continuous fuel-level monitoring with a validated accuracy of ±5

2. **Cloud Based Data Transmission**

   a) Implementation of GSM-based communication protocols to transmit fuel-level and GPS data from the vehicle to a centralized cloud server with a latency not exceeding 3 seconds.

3. **GPS Enabled Route Tracking**

   a) Incorporation of GPS modules for logging vehicle location at a sampling rate of 5 seconds, allowing correlation of fuel consumption with route patterns and driving behavior.

4. **Anomaly Detection via Machine Learning**

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a) Context-aware fuel deviation detection model development by means of unsupervised learning methods (e.g., Isolation Forest, Matrix Profile).

b) Analysis of precision and false positive rate of the model in a controlled setting.

5. **Driver Rest Alerts and Maintenance Reminders**

a) Incorporation of a dashboard system deployed at the trucking company's base to show cumulative fuel and distance measures.

b) Production of automated rest alerts (e.g. every 300 km or after 6 hours) and maintenance reminders (e.g. after every 5,000 km for checking oil).

1.7.3 Delimitations

1. **Focus on Fuel Monitoring and Anomaly Detection**

    - The system will be designed to focus specifically on fuel level monitoring, detecting consumption anomalies, and providing maintenance guidance based on fuel usage and distance traveled. It will not incorporate other real-time diagnostic data from the vehicle's onboard systems.

2. **Hardware and Communication Constraints**

    - The system utilizes digitally calibrated float sensors, an embedded GPS module, and a microcontroller interfaced through the truck's ECU/diagnostic connector; higher-end industrial sensing platforms are not employed.
    - Real-time data transmission primarily uses IoT communication protocols, enabling low-latency delivery of fuel and GPS data to the cloud backend.

- A cellular module (LTE & GSM) functions as a fallback communication pathway, activated only during LoRa connectivity loss, and is limited to transmitting essential alerts and critical status messages due to bandwidth and cost constraints.
- Other wireless communication technologies beyond the selected IoT modules are not implemented within the system.

3. **Limited Fleet Size and Geography**

- The system testing will be limited to the number of vehicles provided for use by the trucking company within a specific geographic area. Large scale testing and deployment across a wider range of fleet types and terrains are not covered.

4. **User Features**

- The cloud based dashboard is designed for the fleet managers use only and does not support interaction from the drivers.
- The cloud based dashboard is limited to essential features such as real time fuel level monitoring, GPS tracking, driver rest alerts, and maintenance reminders based on distance and fuel data. Advanced analytics features are not part of the current dashboard plans.

1.8 Description and Methodology

This study employs a quantitative research paradigm combined with a data-centric, system development approach to implement an AI-driven fleet fuel monitoring and anomaly detection system. The methodology integrates theoretical foundations from Chapter 3 on cyber-physical systems and IoT architectures, system design principles from Chapter 4, and practical implementation strategies detailed in Chapter 5. The system leverages real-time sensor data, including digitally calibrated float sensors interfaced via the vehicle's CAN bus or OBD-II port, GPS modules for location tracking, and embedded alert subsystems for driver safety. These components feed into a microcontroller that preprocesses signals, synchronizes time-series data, and transmits information to a cloud platform primarily via LoRa, with GSM as a backup channel.

Offline, historical and hybrid datasets, combining fleet telemetry and publicly available vehicle logs, are used to train non-parametric unsupervised machine learning models such as Matrix Profile and Isolation Forest. These models identify anomalous fuel consumption patterns and provide context-aware alerts. In the cloud, a Python-based engine continuously evaluates incoming data, logs anomaly events, and stores synchronized measurements in a centralized database. Users access the information through a web-based dashboard that displays real-time truck status, historical fuel usage, route playback, and operational alerts.

The methodology includes signal preprocessing (smoothing via moving averages), time synchronization using linear interpolation, hybrid dataset construction, and rigorous evaluation metrics (precision, false positive rate, accuracy, detection latency) to validate system objectives, as elaborated in Chapter 5. The approach follows iterative, agile development cycles, allowing continuous refinement of sensor integration, communication protocols, and anomaly detection algorithms, thereby ensuring reliability, scalability, and actionable fleet monitoring insights.

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1.9 Research Dissemination Plan

The findings of this study will be disseminated to relevant academic, industrial, and technical stakeholders to promote the adoption and further development of intelligent fleet monitoring technologies. Upon approval of the study, the final manuscript will be submitted to the institutional repository of De La Salle University, including the Animo Repository, in accordance with university research archiving policies. Primary target audiences include the research partner Hino Paranaque Subic GS Auto Inc. (SGSA), fleet managers, logistics operators, academic researchers, engineering students, and faculty members within the field of IoT, transportation systems, and machine learning.

The dissemination process will occur in multiple stages. During the implementation and testing phase, progress updates and preliminary findings will be presented to SGSA representatives and thesis advisers for technical evaluation and operational feedback. Upon completion of the study, the final results, including system performance metrics, anomaly detection accuracy, latency measurements, and operational observations, will be formally presented through the thesis defense at De La Salle University. A digital and printed copy of the approved manuscript will also be submitted to the institutional repository of De La Salle University, Animo Repository, in accordance with institutional requirements. In addition, the proponents may present selected findings in future academic conferences, research exhibits, or technology demonstrations related to IoT systems, intelligent transportation, or artificial intelligence applications.

The dissemination timeline will follow the project completion schedule presented in Section 1.9, including the individual Gantt charts of the proponents. Preliminary technical findings and prototype demonstrations will be shared during the system testing and validation phase. Final dissemination activities, including the thesis defense, manuscript submission, and partner presentation, will occur after the completion of data analysis, evaluation, and manuscript revisions. Any future publication or presentation derived from the study will only use anonymized and non-sensitive operational data to preserve confidentiality and comply with ethical and data privacy standards.

The effectiveness of the dissemination activities will be evaluated through several criteria, including feedback from thesis panel members, adviser evaluations, partner company assessments, and audience engagement during presentations and demonstrations. Additional indicators of dissemination effectiveness include successful acceptance of the manuscript, utilization of recommendations by the partner company, and the potential application or continuation of the proposed system in future research or operational deployments.

1.10 Conflict of Interest

None.

1.11 Estimated Work Schedule and Budget

1.11.1 Gantt Chart

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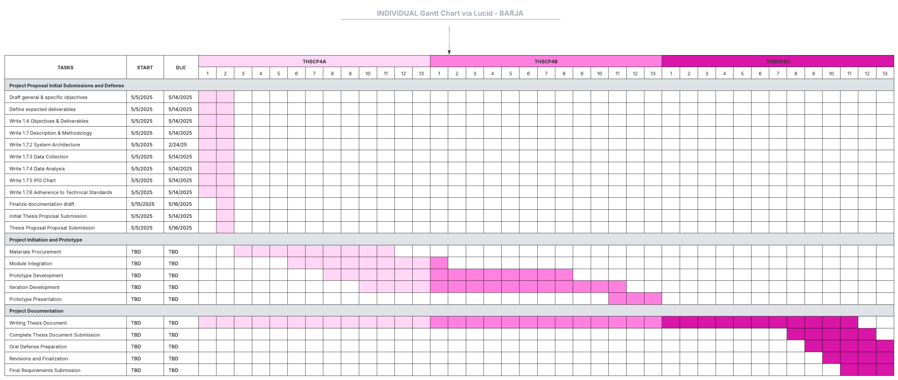

Fig. 1.1 Individul Gantt Chart for Barja

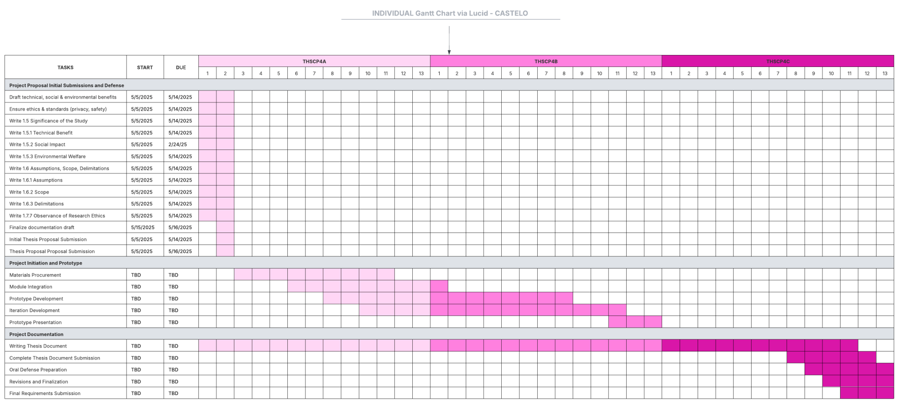

Fig. 1.2 Individul Gantt Chart for Castelo

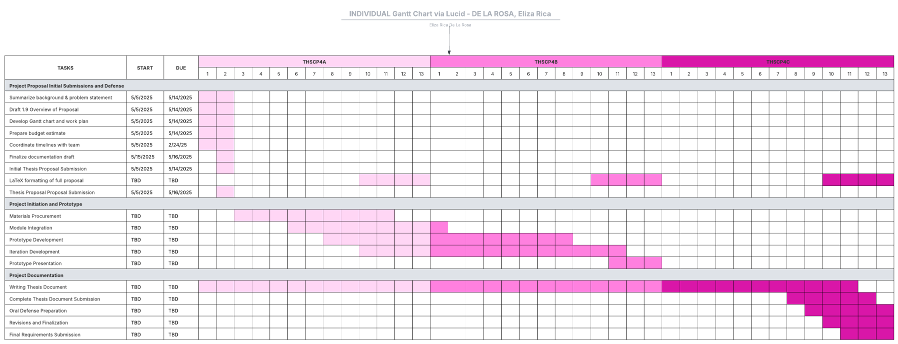

Fig. 1.3 Individul Gantt Chart for De La Rosa

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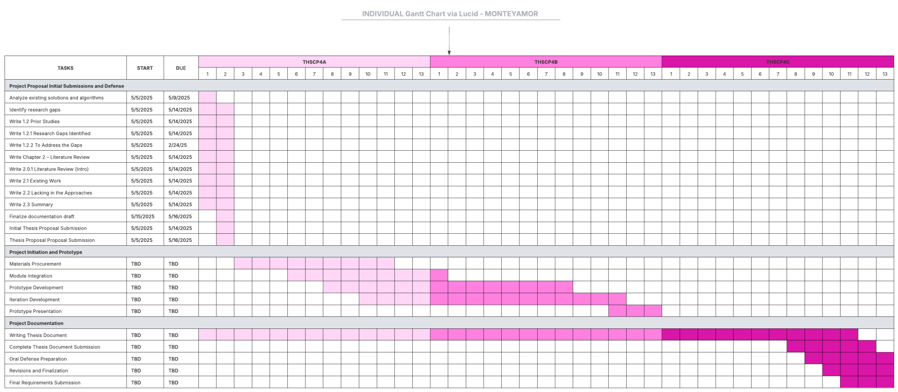

Fig. 1.4 Individul Gantt Chart for Montemayor

1.11.2 Estimated Budget

TABLE 1.3 DETAILED BUDGET BREAKDOWN FOR THE PROJECT

Item / Activity	Unit	Qty	Unit Cost (PHP)	Total Cost (PHP)	Remarks
A. Covered by Hino Motors					
Use of truck for testing	Truck	2	Sponsored	-	Provided by Hino
B. Covered by Research Team					
Hardware Components					
CAN Bus / OBD-II Interface Module		1	1,200	1,200	Reads fuel levels and vehicle parameters from ECU
GPS Module	-	1	650	650	Route tracking and speed monitoring
ESP32 DevKit 30 pins	-	2	350	700	Main microcontroller boards
A7670E	-	1	1,300	1,300	Cellular communication module
MCP2515 CAN Bus Module	-	1	91	91	CAN bus interface module
Buzzer Module	-	1	45	45	Audio alert component
Buttons	-	2	10	20	Driver input controls
Arduino TFT Screen 3.5 in	-	1	648	648	In-vehicle display interface
LoRa Module Ra-02 433 MHz	-	2	218	436	Long-range communication modules
OBD2 16-pin connector	-	1	154	154	Vehicle diagnostic connector
GPS Module NEO-M8N	-	1	600	600	GPS tracking module
Resistors (1x220 ohms, 1x5 kohms)	-	2	1	2	Circuit protection components
SIM card (TNT)	-	1	100	100	Cellular network access
Prototype Case					
3D Model and Printing	-	1	4,500	4,500	Prototype enclosure fabrication
Total Estimated Budget				**10,446**	

1.12 Overview of the Thesis Proposal

This chapter provides an overview of the study, which addresses inefficiencies in logistics and trucking operations while tackling safety concerns related to fuel monitoring, vehicle tracking, and anomaly detection. Modern trucking operations involve complex coordination, and current approaches—manual fuel measurements, fragmented GPS tracking, and delayed anomaly detection—are insufficient for ensuring operational efficiency and driver safety. This work proposes an integrated, IoT-based intelligent monitoring platform that acquires fuel level through the Toyota meter Mode 22 PID 21/29 channel of the OBD-II interface, with a four-stage filter cascade (median-of-31, EMA, dead-band, direction-aware step rule) implemented on an ESP32-based in-truck HMI, GPS, and cloud-deployed machine learning models to provide real-time insights and actionable alerts.

The problem statement highlights a fragmented monitoring environment with limited automation, reactive maintenance scheduling, and the inability to detect unusual fuel consumption proactively. The objectives of the research focus on developing a real-time system that reads fuel levels from the truck's ECU via CAN bus, tracks location and speed using GPS, and identifies anomalies in consumption patterns using unsupervised machine learning models (Matrix Profile and Isolation Forest). The system is supported by a cloud-based dashboard, which provides fleet managers with fuel usage insights, rest alerts, route tracking, and preventive maintenance notifications. This approach promotes operational efficiency, enhances driver safety, and reduces fuel wastage, making it a technically and environmentally relevant solution.

Building on this foundation, Chapter 2: Literature Review examines previous research on IoT-based fleet monitoring systems, GPS-enabled tracking technologies, and machine learning methods for anomaly detection. The review compares existing methods, identifies gaps in real-time deployment, system integration, and context-aware alerting, and evaluates technology innovations and algorithmic limitations. This analysis establishes the academic basis for the study and demonstrates how the proposed system advances the state of the art in smart fleet management by providing a fully integrated, real-time, and context-aware monitoring solution.

Chapter 2

LITERATURE REVIEW

2.0.1 Literature Review

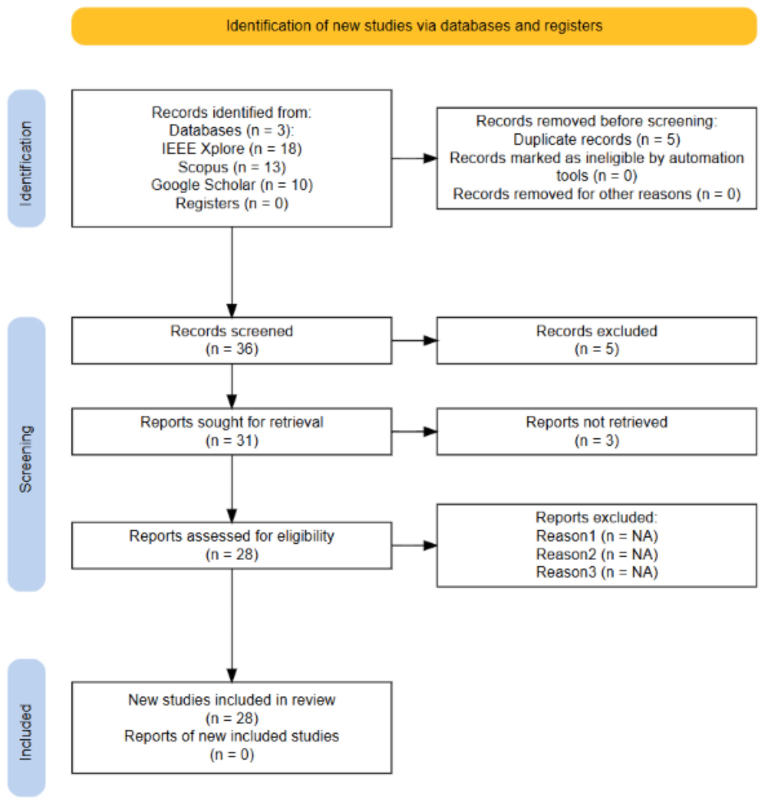

Fig. 2.1 PRISMA Diagram

The PRISMA diagram illustrates the systematic process used for selecting studies for review through screening and identification. The search identified a total of 41 records from three databases: Scopus, IEEE Xplore, and Google Scholar. Five (5) duplicate records were excluded from the 41 records, resulting in 36 records screened. The evaluation of 36 records resulted in five exclusions due to records being outdated/irrelevant, which reduced the number of reports needed for retrieval to 31. Three (3) reports were not retrieved upon retrieval, resulting in 28 reports assessed for eligibility. The final review included 28 studies only, while no new studies were reported.

2.1 Existing Work

2.1.1 Vehicle Tracking and Fuel Sensing Technologies

Fuel Monitoring Systems

Various fuel monitoring systems have been proposed, discussed, and developed in existing studies, emphasizing real-time fuel level, theft indication, and enhanced operational efficiency. In the work of Joshi et al. [Joshi et al., 2024], a system combines ESP32 and Arduino UNO microcontrollers with the ThingSpeak cloud-based platform to measure fuel levels, control transactions, and enable predictive maintenance at fuel stations. Similarly, Ahmed et al. [Ahmed et al., 2017] developed a fuel management system using Raspberry Pi and both ultrasonic and chemical Etape sensors to acquire real-time fuel level data, displayed through a web application.

In the automotive domain, several studies aim to enhance the accuracy of fuel management through sensor integration. The digital system by Sakthimohan et al. [M, 2021] employs ultrasonic and flow sensors with an Arduino UNO to display real-time measurements on an LCD, providing a replacement for imprecise analog gauges. Meanwhile, Farzana et al. [Farzana, 2020] proposed a motion-based fuel monitoring technique that integrates vehicle motion parameters with fuel level sensor readings, achieving 80% success in accurately determining fuel levels and 100% detection success in identifying fuel theft and refills via a smartphone application.

Studies focused on fuel theft detection, such as those by Tatababu et al. [Tatababu et al., 2024] and Mathi et al. [Mathi et al., 2024], utilize capacitive or ultrasonic sensors, GPS tracking, and GSM-based alerts to detect tampering or fuel loss and notify operators immediately. In the Intelligent Fuel Monitoring and Theft Detection System using GPS & GSM, Mathi et al. [Mathi et al., 2024] also monitored battery charge (SoC), with applicability for both hybrid and conventional vehicles, making it suitable for logistics and fleet management.

Ali et al. [Komal Shoukat Ali et al., 2018] presented an automatic fuel tank monitoring, tracking, and theft detection system integrating Arduino-based fuel circuits with GSM/GPS modules and LabVIEW for database management. This system replaces manual fuel tracking with automated alerts and volume monitoring, offering a cost-effective solution for transportation industries. Overall, these systems indicate a trend toward inexpensive, sensor-based, remotely accessible fuel monitoring technologies with high priority on accuracy, security, and real-time data analysis.

GPS and Vehicle Telematics

Global Positioning System (GPS) and vehicle telematics technologies have become essential components in modern vehicle monitoring and fleet management systems. These technologies allow real-time tracking, remote diagnostics, and performance analysis by collecting and reporting data such as location, speed, fuel consumption, and engine status. A common trend in these studies is the combination of GPS and GSM modules to facilitate real-time vehicle tracking, usually complemented by cloud-based or mobile-supported platforms. Joshi et al. [Joshi et al., 2024] proposed a system enabling real-time monitoring for logistic management using sensors, GPS tracking, and cloud-based data analytics. Similarly, Alquhali et al. [Alquhali et al., 2019] and Nada [Nada, 2017] employed GPS and GSM modules to send live location data, show vehicle tracks on electronic maps, and record location history for future reference and analysis. Vigneshwaran et al. [Vigneshwaran et al., 2020] presented a GPS-GSM-based bus tracking system supporting student notifications for bus arrival, emergency notifications, and route alerts, enhancing safety and saving waiting time. The system further offered a framework for real-time fuel monitoring with implications for broader fleet management applications.

2.1.2 LoRa Technology

LoRa (Long Range) is a wireless communication technology engineered for Internet of Things (IoT) and machine-to-machine (M2M) communication. It supports low-power, battery-operated embedded systems that transmit small amounts of data over long distances [Orange Connected Objects & Partnerships, 2016]. Traditional cellular networks such as GSM, 2G, 3G, and 4G prioritize high data throughput, which can result in inefficient power consumption for IoT applications that require infrequent, small data transmissions [Majeed, 2015, Lee et al., 2007]. Unlike cellular networks or short-range protocols such as Wi-Fi and Bluetooth, LoRa is optimized for applications where energy efficiency and communication range are more important than high throughput [Devalal and Karthikeyan, 2018].

The LoRa communication architecture follows a structured flow of information so that small data packets can reach cloud-based systems efficiently:

- **LoRa Node or End Device:** This component includes the sensors or applications where sensing and control take place. It collects telemetry data and transmits them as small, optimized data packets.
- **LoRa Gateway:** The gateway receives packets from end nodes and acts as a bridge between LoRa nodes and the internet.

- **Cloud Backend:** Once the gateway forwards data through a backhaul connection, the information reaches a network or application server where it can be processed, visualized on dashboards, or stored for further analysis.

2.1.3 LoRa Gateway

LoRa gateways are multi-channel, multi-modem transceivers capable of simultaneous reception and decoding of LoRa packets from numerous end nodes. Each gateway acts as a bridge that connects LoRa nodes to the internet by capturing transmitted node data and forwarding them through standard backhaul connections such as Wi-Fi, Ethernet, or cellular data. Data from different gateways are processed through a network server, which functions as the central intelligence of the LoRa architecture. This server performs security authentication, filters messages, manages adaptive data rates, and relays appropriate acknowledgments back to the gateways [Orange Connected Objects & Partnerships, 2016].

Gateway deployment varies according to coverage requirements. Micro gateways are typically deployed in public networks to provide broad city-wide or nationwide connectivity, while pico gateways are used in dense or challenging environments to improve network capacity and service quality. To support these deployment scenarios, LoRa base stations may use omni-directional or multi-sector antenna configurations [Devalal and Karthikeyan, 2018].

2.1.4 Technical Performance Metrics and Environmental Factors

The operational effectiveness of LoRa networks depends on the interaction between transceiver hardware configuration and environmental conditions. According to Chengdu Ebyte Electronic Technology, the communication distance of a LoRa module is influenced by transmit power, receiver sensitivity, and antenna gain [Chengdu Ebyte Electronic Technology Co., Ltd., 2024]. In open rural areas, a standard LoRa layout can achieve communication links of approximately 15 to 20 kilometers. In dense urban environments, however, coverage may decrease to approximately 1 to 5 kilometers because of building penetration losses and multipath propagation effects [Chengdu Ebyte Electronic Technology Co., Ltd., 2024].

Manalu et al. [Manalu et al., 2023] evaluated the performance of a LoRa-based vehicle tracking system using a HopeRF-RFM95W transceiver module operating at 915 MHz on a moving vehicle. The study analyzed link-quality parameters in a Non-Line-of-Sight (N-LoS) suburban environment at speeds of 10 km/h, 20 km/h, and 30 km/h. The evaluation focused on how buildings, vegetation, and other wireless signals affect key performance metrics. Received Signal Strength Indicator (RSSI) measures absolute signal power and degrades behind thick barriers, vegetation, or dense buildings. Signal-to-Noise Ratio (SNR) measures signal clarity against electromagnetic interference and may drop significantly because of disruptions from pedestrians, structures, vehicles, and Wi-Fi towers. Packet loss may also increase sharply near major structural obstacles, showing that LoRa performance in mobile fleet applications must account for physical barriers and N-LoS conditions [Manalu et al., 2023].

2.1.5 Real-World Implementation: LoRa-Based Telemetry and Fleet Management Systems

The long-range and low-power characteristics of LoRa make it suitable for geographic areas with limited commercial telecommunication infrastructure. Common applications include smart agriculture, pasture health monitoring, and isolated industrial operations. However, reliance on a single low-power wide-area network (LPWAN) protocol may introduce communication risks when asset-tracking nodes move between remote rural spaces and dense urban areas.

To reduce blind spots, industrial systems may use hybrid communication architectures. The Meitrack LoRa Fleet Tracking Solution illustrates this approach in large-scale agricultural operations, such as orchards and palm plantations, where GSM coverage may be unreliable or unavailable [Meitrack,]. In such environments, multi-modem tracking hardware can use a dual LoRa and GPRS switching configuration based on network availability. While operating in cellular dead zones, localized sensor payloads, including ultrasonic fuel-level data, driver authentication logs, and safety alerts, can be routed through LoRa to a local base station. This architecture helps preserve asset visibility when standard cellular infrastructure is unavailable [Meitrack,].

2.1.6 GSM and LTE Technologies

The Global System for Mobile Communication (GSM) is a widely recognized standard for digital cellular technology. It originated from a group formed in 1982 to develop a unified European mobile telephone standard for cellular radio systems operating at 900 MHz [The International Engineering Consortium,]. GSM is known for broad geographic coverage and supports voice calls and basic data services through second-generation cellular communication. Long-Term Evolution (LTE), by contrast, is a wireless broadband communication standard that succeeded 3G and became a cornerstone of 4G mobile networks. LTE development began as a Third Generation Partnership Project (3GPP) effort in 2004 to provide a new radio access technology focused on packet-switched data [Miyim and Wakili, 2019].

LTE was developed to address increasing demand for mobile broadband services by improving network capacity, data rates, quality of service, and latency. These improvements support mobile wireless connectivity that aligns with performance benchmarks established by the International Telecommunication Union (ITU). After LTE was established in 2009, further performance requirements led to LTE-Advanced (LTE-A), which satisfied International Mobile Telecommunications (IMT) requirements for fourth-generation networks and achieved peak data rates of up to 1 Gbps [Miyim and Wakili, 2019, Kanchi et al., 2013].

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Fig. 2.2 Globe GSM (pink) and LTE (orange) Coverage Map Philippines (GSMA Intelligence)

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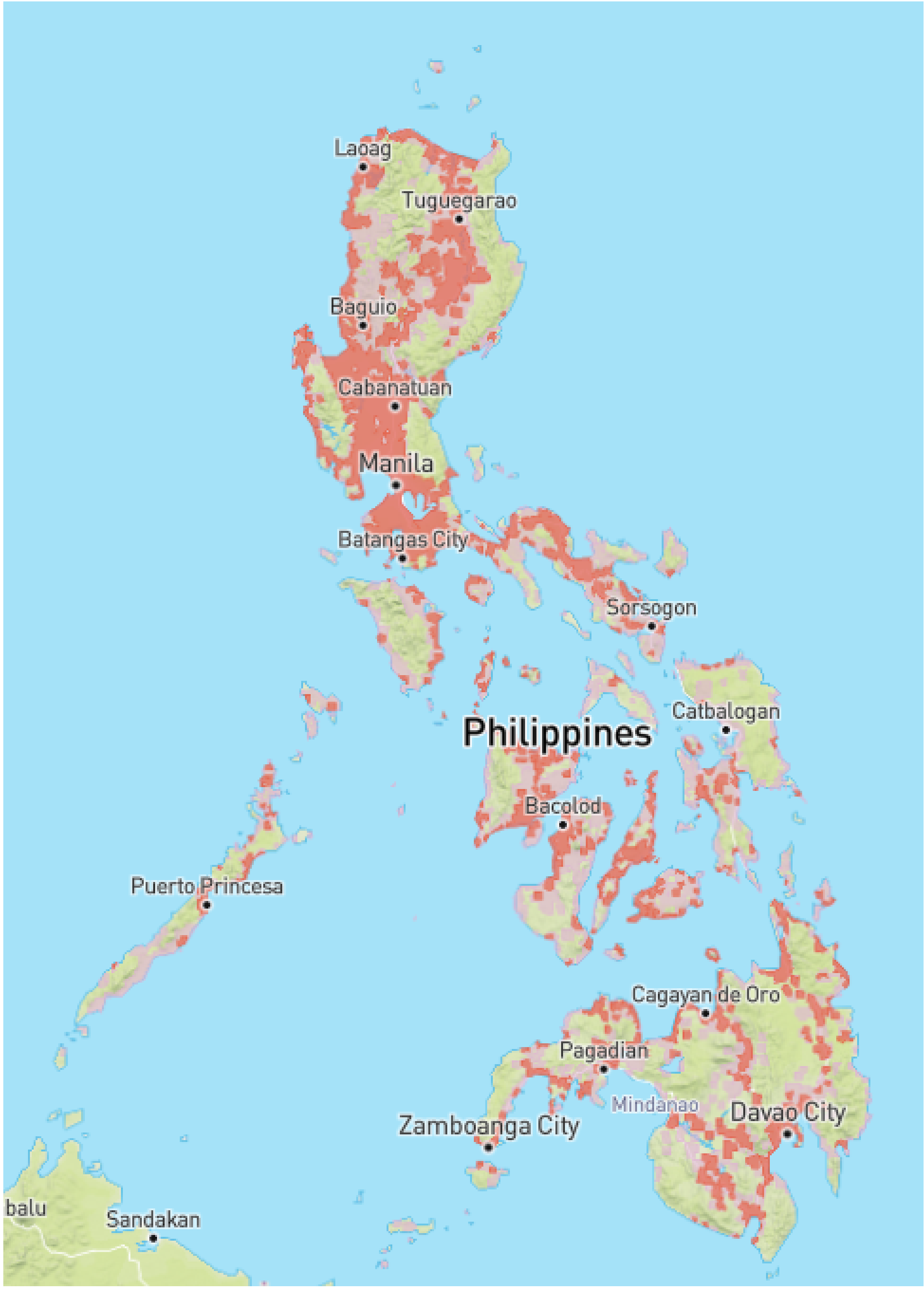

Fig. 2.3 Smart GSM (pink) and LTE (orange) Coverage Map Philippines (GSMA Intelligence)

TABLE 2.1 COMPARISON BETWEEN GSM AND LTE

Feature	**GSM (2G)**	**LTE (4G)**
Network Architecture	Circuit-switched	Packet-switched (All-IP)
Primary Focus	Voice and text services	High-speed mobile broadband
Coverage	Extensive broad geographical reach, including rural and remote areas	High-capacity and density-dependent coverage that requires closer proximity to modern cell towers
Frequency Band	900 MHz and 1800 MHz	Bands 1 to 25 for Frequency Division Duplex and bands 33 to 41 for Time Division Duplex
Signal Propagation	Excellent because of lower frequencies	Variable because higher frequencies generally have shorter range
Infrastructure Needs	Lower base station density	Higher base station density
Data Capability	Low, supporting basic data and messaging	Very high, supporting multimedia and internet access

The coverage maps in Figures 2.2 and 2.3 illustrate GSM and LTE coverage patterns in the Philippines based on GSMA Intelligence coverage data [GSMA, 2026]. Comparing the two networks, Globe shows a noticeably denser and more contiguous spread of both GSM and LTE coverage, particularly through the Visayas and Mindanao, where its colored patches connect more continuously along coastal and lowland routes. Smart's coverage, while still following the same population centers, appears comparatively more fragmented, with more isolated patches and gaps between cities, especially in the southern islands.

In both cases, LTE coverage appears as a smaller subset within the broader GSM footprint. This indicates that 4G service is generally concentrated in and around cities and main highway corridors, while 2G/GSM extends farther into surrounding rural areas as a fallback connectivity layer. This difference is important for fleet monitoring systems because mobile assets may move between high-capacity LTE areas and wider but lower-bandwidth GSM fallback zones [Opensignal, , GSMA, 2026].

The Philippine cellular landscape is transitioning away from legacy networks. In August 2025, the NTC issued Memorandum Circular No. 003-09-2025, mandating phased decommissioning of 2G and 3G services, with nationwide 3G switch-off by 31 December 2026 and 2G on a separate, slower schedule because rural areas still depend on it. The circular also bars new type approval for devices whose primary capability is 2G or 3G. Two design implications follow: LTE must be the primary wide-area path, since it is the network being expanded rather than retired; and any 2G fallback is transitional, which motivates retaining LoRa and on-device buffering where neither LTE nor 2G is dependable [China Academy of Information and Communications Technology, 2025].

2.1.7 Anomaly Detection

Several studies on anomaly detection in vehicular systems aim to improve operational safety, combat theft and fraud, and optimize fuel efficiency. Various technological means, such as GPS tracking, machine learning, and time series analysis, have been used to identify abnormal vehicle behaviors. Crisgar et al. [Crisgar et al., 2021] developed a vehicle tracking system using Google Cloud IoT Core and Firebase to monitor vehicle movements and engine activity for theft detection. Unauthorized ignition or unexpected movement beyond a 50-meter radius is flagged as potential theft, with real-time notifications pushed to a Progressive Web App, enabling users to respond quickly.

Studies focusing on fuel efficiency in heavy-duty vehicles (HDVs) through anomaly detection models include Turan et al. [Turan et al., 2024], who proposed an ensemble of unsupervised machine-learning models (Isolation Forest, Autoencoder, k-NN Regression) combined via weighted majority voting to detect abnormal fuel consumption patterns based on road slope and speed variations. Mumcuoglu et al. [Mumcuoglu et al., 2023] used decision tree ensembles to classify fuel usage as normal or anomalous, achieving 92.2% accuracy and an F1 score of 0.78. Both studies indicate that repeated detection of outliers can highlight anomalies such as mechanical issues or inefficient driving.

Aquize et al. [Aquize et al., 2017] applied Self-Organizing Maps (SOM) to detect abnormal fuel consumption in RFID-registered vehicles in Bolivia. By learning conventional consumption patterns, the SOM model achieved an 80% confidence rate in flagging suspicious activity indicative of fraud. Semenov et al. [Semenov et al., 2024] employed DBSCAN clustering and Local Outlier Factor (LOF) in a three-level telematics platform to forecast and detect truck breakdowns in real-time. This system tracks multiple vehicle parameters, labeling them as normal or outliers based on deviation from learned patterns, minimizing surprise breakdowns and providing early driver alerts. Aquize et al. [Aquize et al., 2017] proposed a fuel anomaly detection framework based on Self-Organizing Maps (SOM) using RFID-registered vehicle data collected from fleet operations in Bolivia. The SOM algorithm learned normal fuel consumption patterns and identified suspicious deviations associated with potential fuel fraud. The study achieved approximately 80% confidence in detecting anomalous fuel usage and demonstrated the effectiveness of unsupervised learning approaches for identifying irregular fuel consumption behavior without requiring labeled datasets.

Similarly, Kaleem Habib and Arif Iqbal Umar [Habib and Umar, 2015] utilized data mining techniques to calculate and detect anomalies in fuel expenses by analyzing historical fuel transaction records and consumption patterns. Their approach identified abnormal expenditure behaviors that could indicate fuel theft, reporting errors, or operational inefficiencies. The findings demonstrate the value of data-driven anomaly detection methods in supporting cost control and fraud prevention within fleet management environments.

Garcia Vega [Garcia Vega, 2022] emphasized time series anomaly detection for vehicle monitoring, developing an iterative approach to detect unusual driving patterns through preprocessing, subsequence segmentation, scoring, clustering, and expert validation. Applied in an industrial context, it identified over 120 validated anomalies, helping diagnose mechanical failures or unsafe driving behavior. The study highlights the importance of visual interpretability tools and recommends future improvements via real-time data processing.

Because published fleet fuel-anomaly work (Barbado and Corcho, 2022) [Barbado and Corcho, 2022] uses boxplot/IQR-based unsupervised detection as an interpretable baseline before applying explainable models, the present study compares Isolation Forest and Matrix Profile against a rule-based / IQR baseline under leave-one-trip-out evaluation.

This baseline serves both as a fairness check (to show that the non-parametric models genuinely add value over a thresholding heuristic) and as an ablation reference in Chapter 5.

2.1.8 Machine Learning Applications

Barbado and Corcho (2022) [Barbado and Corcho, 2022] developed an explainable machine-learning framework for vehicle fuel-consumption anomalies using real-world telematics data from diesel and petrol fleet vehicles. Their study emphasised that anomaly detection should not only identify abnormal fuel consumption but also explain the operational causes behind the anomaly, using XAI techniques on top of an unsupervised detection layer. They first identified candidate anomalies via a univariate boxplot/IQR outlier method and then trained a contextual model that incorporates domain knowledge such as vehicle subgroup, route type, and driver behaviour. This is directly relevant to the present thesis because fuel anomalies in Philippine logistics must be interpreted with operational context - the same context (driver, route, vehicle, fuel-consumption baseline) is encoded in the Isolation Forest Model C feature set used in Chapter 5.

Recent advances in machine learning have been integrated into vehicle systems to improve fuel efficiency, detect anomalies, and enhance operational transparency. In fuel monitoring and anomaly detection, machine learning methodologies extract actionable insights from complex, high-dimensional vehicle data. Fortela et al. [Fortela et al., 2023] applied unsupervised machine learning algorithms (Principal Components Analysis, K-Means Clustering, Self-Organizing Maps) to detect anomalies in fuel economy and emission data from light-duty vehicles (2015–2023). These methods revealed clusters where CO emissions were positively linked to fuel economy, contrary to expected patterns, highlighting irregularities caused by specific fuel test procedures. Barbado and Corcho [Barbado and Corcho, 2022] combined unsupervised anomaly detection with domain knowledge and interpretable machine learning (XAI) to explain potential causes of fuel consumption anomalies. Their models demonstrated up to 88% potential fuel savings on larger datasets and across various user roles.

Barbado and Corcho [Barbado and Corcho, 2022] further demonstrated that interpretable machine learning models such as Explainable Boosting Machines (EBM) can be used not only to detect anomalous fuel consumption but also to explain the factors contributing to those anomalies. Their methodology combines unsupervised anomaly detection, domain knowledge, and explainable artificial intelligence (XAI) metrics to provide actionable recommendations for fleet managers and operators. The study highlights the importance of model transparency in fleet monitoring systems, allowing stakeholders to understand the causes of abnormal fuel usage rather than merely identifying anomalous events.

More recent work sharpens the methodological basis for unsupervised, on-vehicle detection. Cherdo et al. [Cherdo et al., 2023] applied small recurrent and convolutional neural networks to automotive CAN sensor time series, demonstrating that bus data of the kind exposed through an OBD-II interface is amenable to unsupervised anomaly detection, although their models still require gradient-based training. Surveying the domain, Hirth et al. [Hirth et al., 2025] formalize anomaly detection in trucks as identifying deviations across multivariate signals at each time step of a stream, and observe that deep architectures impose retraining and computational burdens that complicate continuous deployment. These observations motivate the two non-parametric methods adopted in this study: Isolation Forest [Liu et al., 2008], which isolates anomalous individual observations without modeling a distribution of normal behavior, and the Matrix Profile [Yeh et al., 2016], which detects anomalous subsequences and has been extended to live data streams [Lu et al., 2022] and to multiple sensor channels [Yeh et al., 2024]. The mathematical formulation of both methods is presented in Chapter 3.

TABLE 2.2 SUMMARY OF EXISTING WORKS

Study	**Technology Used**	**Key Features**	**Strengths**	**Limitations**
Tatababu et al. (2024) [Tatababu et al., 2024]	Ultrasonic/Capacitive sensors, GSM, GPS, Smartphone/Web interface	Theft detection, fuel level alert, real-time remote access	Accurate for remote generator monitoring, GPS-based tracking	Findings specific to the tested generator systems and environments
Joshi et al. (2024) [Joshi et al., 2024]	IoT, GPS, RFID, Cloud Computing	Real-time vehicle/asset tracking, data analytics, cloud-based monitoring	High tracking accuracy, improved logistics efficiency, responsive alert system	Limited by hardware and logistics test case set
Vigneshwaran et al. (2020) [Vigneshwaran et al., 2020]	GPS, GSM	Real-time bus tracking, emergency alert system, communication to students, route updates	Improves safety, scheduling, and passenger information	Not highly scalable; lacks detailed integration with modern telematics platforms
Patil & Patil (2024) [Patil and Patil, 2024]	IoT, GPS, GSM, Accelerometer, IR Sensors	Tracking system with SMS-based communication; remote ignition cut-off mechanism and vehicle password protection	Comprehensive vehicle security, real-time alerts, detection of tampering/impacts	Low adaptability to harsh conditions like GPS/GSM signal loss in remote areas
Fortela et al. (2023) [Fortela et al., 2023]	Unsupervised Machine Learning: PCA, K-Means, SOM	Detects hidden trends and impending anomalies; analyzes correlation between CO emissions and fuel economy	Effectively identifies non-obvious patterns within clusters; highlights flaws in standardized testing procedures	Lacks integration with real-time or operational vehicle data; conclusions depend on historical datasets

Study	Technology Used	Key Features	Strengths	Limitations
Barbado & Corcho (2022) [Barbado and Corcho, 2022]	Unsupervised Anomaly Detection; Explainable Boosting Machine (EBM)	Provides fuel anomaly detection and cause explanations, tailored recommendations for fleet operators	High interpretability via XAI metrics; role-based recommendations; potential fuel savings up to 88%	Performance depends heavily on quality of telematics data or feature relevance
Aquize et al. (2017) [Aquize et al., 2017]	Self-Organizing Maps (SOM)	Detects fraudulent fuel usage based on RFID data patterns	80% accuracy applied to real socio-economic problem	Used only simple SOM model; accuracy can improve with more context variables
Liu et al. (2008) [Liu et al., 2008]	Isolation Forest	Ensemble of isolation trees that identify anomalies through random partitioning	Non-parametric anomaly detection with linear-time complexity and low memory requirements	No labeled data required; superior AUC compared with ORCA and LOF; computationally efficient
Yeh et al. (2016) [Yeh et al., 2016]	Matrix Profile (STAMP)	Z-normalized nearest-neighbor similarity profile for motif and discord discovery	Exact subsequence anomaly detection with minimal parameter tuning	Deterministic, interpretable, and near parameter-free
Lu et al. (2022) [Lu et al., 2022]	DAMP Streaming Matrix Profile	Real-time Matrix Profile computation for high-volume streaming data	Detects discords in continuously arriving data streams at very large scales	Suitable for online and real-time deployment
Yeh et al. (2024) [Yeh et al., 2024]	Multidimensional Matrix Profile	Extension of Matrix Profile for multivariate time-series anomaly detection	Simultaneously analyzes multiple sensor channels	Competitive performance across numerous benchmark datasets and highly interpretable results
Hirth et al. (2025) [Hirth et al., 2025]	Deep Learning-Based Truck Anomaly Detection (Survey)	Comprehensive review and taxonomy of anomaly detection methods for trucks	Defines anomaly detection in multivariate truck telemetry streams	Provides truck-specific insights and identifies current research directions

Study	Technology Used	Key Features	Strengths	Limitations
Cherdo et al. (2023) [Cherdo et al., 2023]	Small RNN and CNN Models	Unsupervised anomaly detection using automotive CAN-bus sensor data	Demonstrates feasibility of on-vehicle anomaly detection under limited hardware resources	Effective for OBD-II/CAN sensor environments and embedded deployment

2.2 Comparison of Commercial Fleet Management Systems

Beyond the academic literature, six widely deployed commercial fleet management platforms were examined to establish the detection methodologies available to operators in practice: Geotab, Samsara, Fleetio, Wialon, Fleet Complete, and Navixy. Each platform's own product documentation was used as primary evidence of its capabilities. Despite mature telematics infrastructure, fuel-loss detection across these systems reduces to fixed or manually configured rules. Samsara's documentation describes its fuel theft detection as a configurable alert on any sudden fuel drop of five percent or more [Samsara, 2026]. Navixy requires operators to configure rate-of-change drain thresholds per sensor and can be set to ignore drains while a vehicle is moving [Navixy, 2026]. Geotab and Fleetio approach the problem from the fuel-card transaction side, flagging exceptions such as purchases exceeding tank capacity or fueling locations inconsistent with vehicle position [Geotab, 2024, Fleetio, 2026]. Wialon detects drain events from fuel-level sensor data and commonly relies on operator review for verification [Wialon, 2026], while Fleet Complete's public documentation centers on fuel usage reporting without specifying an adaptive detection method [Fleet Complete, 2026]. Across all six, no platform learns a vehicle's normal consumption behavior, adapts to operating context, or is designed specifically for Philippine fleet conditions.

TABLE 2.3 COMPARISON OF COMMERCIAL FLEET MANAGEMENT SYSTEMS

System	**Data Sources**	**Fuel Monitoring Method**	**ML for Fuel Anomalies?**	**Detection Methodology**	**Limitations**
Geotab [Geotab, 2024]	OBD/telematics device, fuel cards, GPS	Fuel-up records and engine data reporting	No (rule checks)	Capacity mismatch, purchase greater than capacity, and location discrepancy	Transaction-rule logic; no learned consumption behavior
Samsara [Samsara, 2026]	Telematics gateway, fuel sensor/OBD, fuel cards, GPS	Real-time fuel level and usage reports	No (threshold)	Configurable sudden-drop alert (≥5%)	Static threshold; blind to gradual loss; one rule for all conditions
Fleetio [Fleetio, 2026]	Fuel card transactions and telematics integrations	Transaction-based fuel management	No	Exception alerts for location mismatch and volume greater than tank capacity	Sees only fill-up events; blind between refuels
Wialon [Wialon, 2026]	Fuel level sensors, GPS trackers, optional cameras	Sensor-based fuel charts and drain/refill detection	No	Sensor-threshold drain events with human/video verification	Threshold-based and manually reviewed; reactive
Fleet Complete [Fleet Complete, 2026]	Telematics device, GPS, vehicle data, fuel reports	Fuel usage and fleet reporting	No specified adaptive model	Reporting-based fuel monitoring and exception review	Public documentation does not specify adaptive anomaly detection
Navixy [Navixy, 2026]	Fuel level sensors, OBD/CAN, GPS	Drain/refill event detection from sensor data	No	Per-sensor rate-of-change drain thresholds; option to ignore in-motion drains	Manual tuning; in-motion exclusion can hide moving theft or leaks

2.3 Lacking in the Approaches

While existing technologies in vehicle tracking, fuel sensing, and anomaly detection form the foundation upon which developments can be built, they suffer from critical limitations that prevent their use in real-time, fleet-wide operations. Research works such as Krishnasamy [Krishnasamy, 2019] have employed basic sensor technologies without any validation of fuel measurements in real-time driving environments or estimation of accuracy parameters relative to manual readings. Besides, current systems are limited to specific uses and are not tested for scalability or integration with fleet vehicles like trucks such as institutional/organizational level of motorcycles in the study of RS et al. [RS et al., 2024]. Some works do consider machine learning techniques for anomaly detection, but such techniques are mostly based on static and post-processed datasets, with no real-time on-vehicle analysis. Very few implementations include unsupervised machine learning and advanced context-aware modeling, and precision benchmarks are usually left unreported, if at all addressed.

Across the surveyed commercial systems, including Geotab, Samsara, Fleetio, Wialon, Fleet Complete, and Navixy, several operational capabilities remain insufficiently addressed. First, closed-loop in-cab driver feedback is largely missing because most platforms push alerts to a manager dashboard, whereas this thesis pushes alerts to a driver-facing HMI inside the truck cab with rest/snooze acknowledgments captured in the `alerts` table. Second, offline-tolerant telemetry with explicit replay accounting is rarely documented in academic prototypes or commercial fuel-monitoring systems; this thesis tags every telemetry row with `channel_used`, including `offline_replay`, which lets the backend distinguish live data from replayed data. Third, multi-interface state reconciliation is not typically addressed, while this thesis uses partial unique indexes to enforce one-active-trip-per-truck/driver across three concurrent client surfaces: HMI, mobile, and dashboard.

Ultimately, there is no prior system that provides a full solution that brings together a cloud-connected solution for real-time fuel monitoring with integration of GPS tracking and anomaly detection. Further, it completely lacks features such as automatic rest alerts and predictive maintenance reminders, which are necessary for safe, efficient, and compliant operations. Such shortcomings create an opportunity for a more integrated, intelligent, and real-time solution, and this is what the thesis proposes to address.

TABLE 2.4 IDENTIFIED GAPS, LIMITATIONS, AND PROPOSED SOLUTIONS

Study / Gap	**What they did**	**Limitation**	**How this thesis responds**
Area of precision	Existing models lack context-awareness and show lower precision. Models presented in the study of Barbado & Corcho [Barbado and Corcho, 2022] performed up to 80% of anomalous instances explained.	Prior work does not establish the same precision and false-positive benchmark required for this deployment.	(SO4) To design and evaluate a context-aware fuel usage deviation modeling approach using non-parametric unsupervised machine learning models, like Matrix Profile and Isolation Forest, achieving $\geq$90% precision and $\leq$10% false positives in identifying anomalous fuel consumption patterns within controlled environments.

Study / Gap	What they did	Limitation	How this thesis responds
Barbado & Corcho (2022) [Barbado and Corcho, 2022]	Used real-world telematics and interpretable ML to detect and explain fleet fuel-consumption anomalies; combined box-plot/IQR pre-filtering with contextual XAI models.	Focused on fleet-level analytics and explanation; did not produce an embedded HMI + OBD-II + multi-channel-communication + live-alert prototype that closes the loop in the vehicle cab.	This thesis integrates OBD-II fuel acquisition (Toyota Mode 22 PID 21/29), embedded HMI, GPS tracking, four-channel communication fallback, backend dashboard, mobile driver web app, and unsupervised anomaly detection (IF Model C + Matrix Profile) in one working prototype evaluated under LOTO.
Absence of a comprehensive system integrating fuel anomaly detection with GPS-based tracking, and predictive maintenance and alerts	Previous works, including those by Joshi et al. [Joshi et al., 2024] and Nada [Nada, 2017], focus on real-time tracking using GPS/GSM.	These works lack integration with anomaly detection methods for analyzing fuel consumption patterns or predicting vehicle maintenance issues.	(SO3, SO4, SO5) To capture and log GPS location data via an embedded module, correlating route information with fuel and time data for behavior tracking; design and evaluate a context-aware fuel usage deviation modeling approach using non-parametric unsupervised machine learning models; develop and integrate rest alerts and maintenance reminders into a cloud-based dashboard.

Study / Gap	What they did	Limitation	How this thesis responds
No Philippine transportation context	Reviewed studies and platforms do not model local conditions such as Metro Manila congestion, Port of Manila utilization, the 2014 Manila truck ban, and fuel fraud costs [Japan International Cooperation Agency, 2025, Patalinghug et al., 2015a, Llanto, 2017, Asian Development Bank, 2016].	Models may misclassify congestion-shaped idling and consumption as anomalies when Philippine operating conditions are absent.	(GO, SO4) Collect and train on operational data from the partner fleet, Hino Paranaque Subic GS Auto Inc., so congestion-shaped idling and consumption are learned as baseline behavior.
One detection rule for all trucks	Commercial systems use single fleet-wide settings [Samsara, 2026] or manual per-sensor tuning [Navixy, 2026]; prior studies were validated on one vehicle class, generators [Tatababu et al., 2024], or motorcycles [RS et al., 2024].	Shared thresholding cannot represent each truck's normal consumption profile.	(SO4) Learn per-vehicle baselines that update from each truck's own operating history, removing dependence on a shared manual threshold.
Detection of sudden drops only	Threshold and transaction logic are commonly used to detect sudden losses [Samsara, 2026, Fleetio, 2026]; Navixy can ignore in-motion drains [Navixy, 2026].	These approaches may miss slow siphoning and gradual leaks.	(SO4) Apply the Matrix Profile for anomalous-subsequence or discord detection of gradual loss, paired with Isolation Forest for sudden point anomalies.
No real-time analytics or live alerts	Machine learning studies were validated offline [Fortela et al., 2023, Barbado and Corcho, 2022]; deep models burden streams [Hirth et al., 2025, Cherdo et al., 2023]; and Wialon relies on human review [Wialon, 2026].	Prior approaches do not provide live anomaly scoring and alert delivery across operational interfaces.	(SO2, SO5) Deploy a cloud pipeline scoring live OBD-II/GPS streams with Isolation Forest and a streaming Matrix Profile [Lu et al., 2022], issuing live alerts to the dashboard, HMI, and mobile application.

2.4 Summary

With this chapter, it was discovered that there have been various studies on vehicle tracking, fuel sensing, and anomaly detection. The PRISMA diagram shows the systematic approach that was used in the identification of the studies, out of which the 41 initial records were narrowed down to 28 after excluding the duplicates, outdated/irrelevant, and not retrieved records. Moreover, the existing studies presented that though a lot has been done to develop individual systems for fuel level monitoring, GPS tracking, and anomaly detection, most of these works are either narrowly focused or operate in isolation. Many rely on basic sensor integrations without advanced analytics, or implement machine learning models without connecting them to real-time, in-vehicle data streams.

Numerous studies have exhibited the application of different sensor-based systems for fuel monitoring in real-time and detecting theft, in many cases involving microcontrollers, GSM, and GPS modules. Likewise, research on vehicle telematics illustrates the application of GPS in route tracking and fleet management. In the field of anomaly detection, some methods based on unsupervised machine learning and clustering techniques have been utilized to detect abnormal fuel consumption patterns and flag fraudulent or inefficient activity. Yet, most of these researches are based on special scenarios, are not integrated with real-time operational data, or depend on historical data.

While advances have been made in each of these domains, a significant void exists in creating a holistic, real-time solution that integrates fuel level measurement, GPS tracking, and intelligent anomaly detection through machine learning. Current systems exist in silos, are vehicle type- or use case-specific, and do not commonly incorporate predictive maintenance or driver feedback mechanisms.

This research seeks to fill these gaps by designing and testing an integrated IoT-based system tailored for trucks. The system will be an integrated system that combines real-time fuel monitoring, GPS tracking, and advanced anomaly detection using machine learning which aims to improve operational efficiency, driver safety, and maintenance planning.

Chapter 3

THEORETICAL CONSIDERATIONS

This chapter establishes the theoretical foundations underpinning the proposed IoT-based fleet fuel monitoring and anomaly detection system. It presents the essential concepts, mathematical models, and scientific principles that guide the development of the system. These theories act as the main foundation of the study, supporting the methodological decisions described in Chapter 5. By understanding the underlying theories—ranging from cyber-physical systems to signal processing and machine learning—the relevance and rigor of the chosen approach become clear.

3.1 Overview

The methodologies used in the system are grounded in established theories in embedded systems, time-series processing, IoT architectures, unsupervised machine learning, and vehicular telematics. These theoretical concepts justify the design choices made in the development of real-time fuel monitoring components, GPS-based tracking, and anomaly detection models. This chapter explains the foundations behind the signal models, communication processes, and machine learning algorithms that make accurate and robust system performance possible.

3.2 Theoretical Foundations

3.2.1 Cyber-Physical Systems (CPS)

The proposed solution functions as a Cyber-Physical System, integrating embedded sensors, microcontroller-based processing, cloud computation, and real-time communication. CPS theory emphasizes:

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- the tight coupling between physical signals and digital processing,
- deterministic timing and synchronized data flow,
- continuous monitoring and computational feedback loops.

These principles justify the system's architecture, where fuel-level signals, GPS data, and operational states are processed and transmitted in real time.

3.2.2 Sensor Theory and Signal Measurement

Fuel-level readings extracted through the CAN bus/OBD-II interface follow digital measurement theory. Sensor signals can exhibit noise, quantization error, and environmental fluctuation. To improve stability, the system uses a Simple Moving Average (SMA) filter, a standard smoothing method defined as:

$$x_t^{\text{smooth}} = \frac{1}{N} \sum_{i=0}^{N-1} x_{t-i} \tag{3.1}$$

This reduces high-frequency noise and creates a clearer trend for anomaly detection algorithms.

3.2.3 OBD-II and CAN-based Vehicle Data Acquisition

Vehicle data can be acquired through a standardised diagnostic interface rather than an added in-tank sensor. The underlying transport is the Controller Area Network (CAN), a message-based broadcast bus standardised in ISO 11898 in which contention is resolved by non-destructive bitwise arbitration: priority is encoded in the message identifier, and the highest-priority frame proceeds without loss of data or time. Standard CAN uses 11-bit identifiers and extended CAN uses 29-bit identifiers.

On-Board Diagnostics II (OBD-II) operates over CAN as a request–response protocol, with its application layer defined by SAE J1979 / ISO 15031-5 and multi-frame transport by ISO 15765-4. It distinguishes standardised current-data PIDs (Mode 01), which are mandated and uniform across manufacturers, from enhanced PIDs (Mode 22), which are manufacturer-specific and use proprietary scaling. Because Mode 22 scaling is not publicly standardised, any value read through it must be validated empirically, introducing measurement uncertainty.

Heavy-duty commercial vehicles conventionally use SAE J1939, a higher-layer protocol over 29-bit CAN identifiers in which Parameter Group Numbers (PGNs) group signals and Suspect Parameter Numbers (SPNs) define each signal's bit position, scaling, and units. Under J1939, fuel level is a standardised SPN broadcast periodically without an explicit request. The theoretical contrast is therefore between a polled, proprietary request–response model (OBD-II Mode 22) and a push-based, standardised broadcast model (J1939): the former trades standardisation and immediacy for broad device compatibility. Analog fuel senders accessed through either path are subject to slosh-induced fluctuation and quantisation error, which motivates the filtering theory developed next [International Organization for Standardization, 2015, SAE International, 2023, International Organization for Standardization, 2021, SAE International, 2016, SAE International, 2017, International Organization for Standardization, 2024, Robert Bosch GmbH, 1991].

3.2.4 Signal Filtering for Fuel-Level Stabilisation

Raw signals from physical sensors carry impulse noise, low-frequency disturbance, and quantisation error; no single filter addresses all three, so a multi-stage cascade is used, each stage grounded in distinct theory.

Median filter. A running median is a non-linear order-statistic filter that replaces each sample with the median of a sliding window of length W:

$$x_t^{\mathrm{med}} = \mathrm{median}(x_{t-W+1}, \ldots, x_t). \tag{3.2}$$

Its breakdown point is 50%; up to half the window may be corrupted by impulse noise before the output is affected, whereas the arithmetic mean has a 0% breakdown point.

Exponential moving average. An EMA is a first-order recursive IIR low-pass filter:

$$y_t = \alpha x_t^{\mathrm{med}} + (1-\alpha) y_{t-1}, \qquad 0 < \alpha \le 1, \tag{3.3}$$

in which past samples decay geometrically and the smoothing factor α sets a time constant $\tau = -1/\ln(1-\alpha)$; a smaller α gives heavier smoothing at the cost of responsiveness. The simple moving average, by contrast, is a finite-impulse-response filter optimal for white noise but without impulse robustness and requiring the full window in memory.

Dead-band (hysteresis). A dead-band publisher emits a new value only when the smoothed signal departs from the last published value by more than a threshold:

$$\mathrm{publish}(y_t) \Leftrightarrow |y_t - y_{\mathrm{pub}}| \ge \delta. \tag{3.4}$$

Direction-aware gating. An asymmetric step rule treats increases and decreases differently according to the physical behaviour of the measured quantity, and a warm-up guard suppresses output until the window is fully populated, since earlier samples are unrepresentative.

3.2.5 Classification and Evaluation Metric Theory

Detectors are evaluated as binary classifiers. From the confusion matrix (TP, FP, TN, FN):

$$\mathrm{Precision} = \frac{TP}{TP + FP} \tag{3.5}$$

$$\text{FPR} = \frac{FP}{FP + TN} \tag{3.6}$$

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3.7}$$

$$L_d = T_{\text{detected}} - T_{\text{actual}}. \tag{3.8}$$

Accuracy is uninformative under class imbalance, where it is dominated by the majority class [Powers, 2020]. A Receiver Operating Characteristic (ROC) curve plots true-positive rate against false-positive rate, and its area equals the probability that a random positive outranks a random negative [Fawcett, 2006]. Under heavy imbalance, however, ROC-AUC is optimistic; the Precision–Recall curve is more informative, its area collapsing toward the positive prevalence so that a modest value reflects the base rate rather than poor detection [Davis and Goadrich, 2006, Saito and Rehmsmeier, 2015]. Detection latency is $L_{\text{det}} = T_{\text{detected}} - T_{\text{actual}}$, where T_{actual} is when the anomaly occurred and T_{detected} is when it was flagged; the Numenta Anomaly Benchmark scores latency with a time-decaying reward [Lavin and Ahmad, 2015]. For binomial proportions estimated from few positives, the Wilson score interval is preferred over the normal-approximation (Wald) interval [Wilson, 1927]:

$$\frac{\hat{p} + z^2/(2n) \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + z^2/n}. \tag{3.9}$$

Leave-one-group-out cross-validation, here leave-one-trip-out, holds out each group entirely to prevent leakage from within-group correlation [Arlot and Celisse, 2010]. Finally, evaluation against injected synthetic anomalies provides exact ground truth but bounds performance on the injected classes only, since these may not represent the full distribution of real events [Chandola et al., 2009].

3.2.6 Time-Series Synchronization Theory

Sensor streams often arrive at irregular timestamps. Synchronizing them requires interpolation theory. For two samples at timestamps T_i and T_{i+1}, the interpolated point at t_j is computed as:

$$x_j = x_i + \left(\frac{x_{i+1} - x_i}{T_{i+1} - T_i} \right) (t_j - T_i) \tag{3.10}$$

This ensures aligned and consistent time-series data across fuel level, GPS speed, and distance.

3.2.7 IoT Communication Theory

The communication subsystem relies on Internet of Things (IoT) principles and long-range wireless communication theory. LoRa operates using Chirp Spread Spectrum (CSS) modulation, characterized by:

- long-range transmission,
- low power consumption,
- robustness to interference.

Theoretical latency between sender and cloud receiver is expressed as:

$$L = T_{\text{received}} - T_{\text{sent}} \tag{3.11}$$

Understanding latency is essential for analyzing responsiveness and real-time performance.

3.2.8 GPS Temporal and Spatial Theory

GPS logging follows Global Navigation Satellite System (GNSS) principles. To evaluate consistency of GPS sampling, the inter-arrival interval is computed as:

$$t_i = T_i - T_{i-1} \tag{3.12}$$

Maintaining a stable interval ensures accurate reconstruction of routes and distance traveled.

3.2.9 Unsupervised Anomaly Detection Theory

The anomaly detection component relies on non-parametric, unsupervised machine learning, which identifies unusual behavior in multivariate time-series data without labeled training samples. Following the taxonomy adopted for vehicle anomaly detection [Hirth et al., 2025], three anomaly types are distinguished: point anomalies, or individual observations that are anomalous with respect to the data, such as a sudden fuel drop while stationary; contextual anomalies, which are anomalous only in context, such as high consumption at zero speed; and collective or subsequence anomalies, where a sequence of individually unremarkable observations forms an anomalous pattern, such as gradual siphoning. The system assigns point and contextual anomalies to Isolation Forest, operating on engineered contextual features, and collective anomalies to the Matrix Profile.

3.2.9.1 Matrix Profile Theory

Matrix Profile computes similarity between subsequences within a time series. For a time series T of length n and window size m, the profile value is:

$$P[i] = \min_{j \neq i} \|T_{i:i+m-1} - T_{j:j+m-1}\| \tag{3.13}$$

The subsequences with the highest $P[i]$ values, known as discords, are likely anomalies. This method is ideal for detecting sudden fuel drops or siphoning patterns.

3.2.9.2 Isolation Forest Theory

Isolation Forest isolates anomalies through random partitioning. An anomaly score is expressed as:

$$s(x) = 2^{-\frac{E(h(x))}{c(n)}} \tag{3.14}$$

Where:

- $E(h(x))$ is the expected path length for observation x,
- $c(n)$ normalizes the tree depth by dataset size.

Higher scores indicate higher likelihood of anomalous fuel behavior.

3.3 Summary

This chapter presented the theoretical foundation for the system's methodology. It discussed the key concepts supporting real-time sensing, signal preprocessing, GPS logging, IoT communication, and unsupervised anomaly detection. By establishing these theoretical pillars, this chapter clarifies the scientific basis that makes the system's design viable, accurate, and aligned with established models in engineering and data science.

Chapter 4

DESIGN CONSIDERATIONS

This chapter presents the design considerations that guided the development of the proposed IoT- and AI-driven fuel monitoring and anomaly detection system. Whereas Chapter 3 established the theoretical foundations of IoT architectures, cyber-physical systems, signal processing, and unsupervised anomaly detection, the present chapter translates those theories into practical engineering decisions that were directly implemented in Chapter 5.

These design considerations serve as the structural "columns" of the system: grounded in accepted standards, ethical guidelines, and engineering best practices, but tailored to support the procedural steps carried out in Chapter 5. The discussion begins with a comprehensive account of the technical standards that ensure reliability, safety, interoperability, and data security, followed by software and hardware design decisions. Ethical considerations—including privacy, responsible AI, and driver safety—are also presented.

4.1 Standards

4.1.1 Adherence to Technical Standards

To ensure reliability, accuracy, and safety, the proposed fuel anomaly detection system is developed in accordance with established engineering and IT standards. These standards ensure interoperability, scalability, and long-term viability. They directly influence the system architecture, data handling, and model implementation workflows outlined in Chapter 5.

1. **Standards for Sensors and Hardware**

   The system integrates sensors and modules that employ standard communication protocols—**CAN bus/OBD-II, UART, I2C, and GPIO**—consistent with IEEE 1451 smart transducer guidelines [IEEE Standards Association,]. The CAN bus/OBD-II fuel data acquisition aligns with industry telematics conventions [iso, 2020], while the GPS module adheres to global navigation and timing standards [NASA EarthData, 2025].

Safety-related components (buzzer, indicators, MCU GPIO outputs) comply with IEC 61131-2 and ISO 7637 automotive power and electromagnetic compatibility guidelines [iec, , iso, d]. All embedded electronics meet CE and FCC certifications [cec, 2022, fcc,] to ensure environmental and electrical safety.

2. **Data Acquisition and Signal Processing**

   The fuel-level signal is stabilised by a multi-stage filter, a median stage for impulse rejection followed by an exponential moving average for smoothing, rather than a single moving average.

   Time alignment of fuel, GPS, speed, and engine readings follows ISO/IEC 19510 principles for structured time-series data [iso, e]. These standard-aligned preprocessing steps feed into the anomaly detection models described in Chapter 5.

3. **Communication and Data Transmission**

   Communication follows TCP/IP standards under IETF and IEEE 802.11 [rfc, , Samainstage, 2025], while MQTT serves as the primary IoT protocol following OASIS specifications. End-to-end encryption via TLS/SSL aligns with ISO/IEC 27001 cybersecurity requirements [iso, 2022a].

These communication strategies form the basis for the system's message-handling and cloud-upload mechanisms in Chapter 5.

4. **Machine Learning and Model Validation**

   The Matrix Profile and Isolation Forest models are implemented using open-source libraries, and their evaluation follows ISO/IEC 29119 software quality standards [iso, 2022b]. Validation metrics—confusion matrices, ROC curves, precision–recall graphs—provide statistical grounding for model evaluation, as presented in Chapter 5.

5. **Software Architecture and Data Management**

   The modular software design complies with ISO/IEC/IEEE 42010 architectural principles [iso, g]. ACID-compliant storage and GDPR-/ISO 27018-aligned data protection protocols [iso, f] ensure data integrity and privacy. All processed datasets follow FAIR principles.

4.1.2 Observance of Research Ethics

Ethical considerations guide the responsible deployment of the system, especially given the safety-critical and privacy-sensitive context of transportation fleets. Three major ethical domains were identified: data privacy, system safety, and responsible AI.

Data Privacy and Security

The system adopts:

- informed consent prior to any data collection,
- strict access control and secure data storage,

- anonymization and aggregation when feasible,
- transparency in data usage and access rights,
- compliance with national and international privacy regulations.

System Safety and Occupational Risk Management

Given the operational risks of fleet environments, the design aligns with ISO 45001, ISO 26262, and ISO 39001 [iso, c, iso, a, iso, b]. Safety-oriented measures include:

- minimizing driver distraction,
- reducing false alarms through validation,
- providing adequate training for users,
- implementing clear protocols for anomaly response.

Responsible AI and Algorithmic Accountability

Consistent with IEEE P7003 [iee,], the system incorporates:

- explainability mechanisms for anomaly outputs,
- documentation of roles and responsibilities in alert handling,
- defined liability procedures for system errors,
- safeguards against algorithmic bias.

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4.2 Technical Design Considerations

This section elaborates on the engineering decisions and standards that guided the development of the proposed IoT- and AI-driven fuel monitoring and anomaly detection system. Each component, software layer, algorithm, and communication protocol is grounded in established standards to ensure reliability, safety, interoperability, and security. The references indicate where each design decision is applied in the implementation procedures described in Chapter 5.

TABLE 4.1 FINAL HARDWARE AND SUBSYSTEM DECISION TABLE

Subsystem	**Final choice**	**Alternative considered**	**Reason for final choice**	**Known limitation**
Fuel acquisition	OBD-II Toyota Mode 22 PID 21/29	External capacitive / float tank sensor	Non-invasive, no tank modification, ethically clearable, partner-acceptable	Manufacturer dashboard gauge used as field reference (no dipstick validation)
Controller	ESP32 (dual-core, 240 MHz, WiFi+BT)	Arduino Uno / Raspberry Pi	Adequate GPIO + WiFi + dual core for IDF VSPI separation; PlatformIO ecosystem; FreeRTOS scheduling	Limited RAM for on-device ML; ML inference moved to backend
HMI display	ILI9341 2.4” TFT + XPT2046 touch + dedicated buttons	OLED + scroll button only	Full route / fuel / alert UI in-cab; suitable readability in daylight	Touch only used for non-critical actions; physical buttons used for safety-critical confirm
GPS	Embedded GPS (Neo-style) + mobile-app GPS fallback fused at backend	Dashboard-only / external assist only	Resilient to embedded-GPS cold start or building occlusion; tagged via `gps_source` column	Urban-canyon drift; cold start of embedded module on first power-on

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Subsystem	Final choice	Alternative considered	Reason for final choice	Known limitation
Wide-area comms - primary	LTE via A7670E modem	GSM-only baseline	Higher throughput, lower per-packet latency once attached	HTTPACTION + TLS renegotiation can push median latency to approximately 23–24 s on certain Smart cells
Wide-area comms - fallback	GSM 2G via same A7670E (with CGNAPN/CFUN+CSTT fallback)	LTE-only	Retained for legacy 2G-only coverage on some Batangas / Tagaytay routes	Lower throughput; not all carriers maintain 2G
Local-area / depot comms	LoRa SX1278 @ 433 MHz, SF7, BW 125 kHz, 17 dBm	WiFi-only at depot	Long range, low power, bidirectional fleet-yard polling without LTE cost	Low duty-cycle limits payload size; 125 kHz BW shared with other depot nodes
Hotspot fallback	WiFi (driver phone tether or depot AP)	No fallback	Recovers comms when LTE+GSM+LoRa are all down but a personal hotspot is available	Depends on driver compliance to enable hotspot
Offline buffer	SPIFFS append-only ring buffer, replayed with `channel_used='offline_replay'`	RAM-only buffer	Survives reboot; preserves `seq` + `event_id` for backend deduplication	Flash wear if drain is starved for very long periods
Backend	Node/Express on Render + Supabase Postgres	Self-hosted Postgres on a VPS	Managed DB, row-level security, free-tier acceptable for thesis-scale traffic	Render cold-start adds latency to the first POST after idleness

Subsystem	Final choice	Alternative considered	Reason for final choice	Known limitation
ML detector	Isolation Forest Model C (contamination=0.015, n_estimators=300, max_samples=256)	Threshold-only / IQR baseline; Hybrid Union; Hybrid Intersection	Only Isolation Forest variant that met ≥90% precision and ≤10% FPR under LOTO	Recall depends on injection realism; will be retrained as live anomaly corpus grows
Subsequence detector	Matrix Profile (window=12, z-thresh=2.0)	Drop entirely	Useful confirmatory layer for slow-leak / steady-siphon patterns that single-point IF misses	Less robust than Model C at row-level scoring; retained as supporting evidence only

4.2.1 Hardware Selection and Integration

The hardware configuration is designed to support the end-to-end workflow of the system (see Chapter 5.9). The following standards and implementation points apply:

- **MCU:** ESP32/NodeMCU serves as the edge processing device for real-time acquisition and preprocessing of fuel, GPS, and vehicle telemetry data. It complies with IEC 61131-2 for embedded controllers [iec,] and ISO 7637 for automotive electrical transients [iso, d]. In Chapter 5.9, the MCU handles sensor filtering, timestamping, and local anomaly scoring.
- **GPS Module:** The UART-connected GPS module provides geolocation and speed data, adhering to global navigation standards [NASA EarthData, 2025]. Its signals are synchronized with fuel readings during preprocessing in Chapter 5.9.

- **CAN/OBD-II Interface:** The CAN/OBD-II interface was selected as the acquisition layer because it satisfies the non-invasive constraint set by the partner operator. From this one port the prototype reads speed, RPM, engine load, throttle, MAF, coolant temperature, runtime, DTC status, and the Toyota Mode 22 PID 21/29 fuel-sender voltage encoded at 0.5 L per ADC count. Mode 22 access is manufacturer-specific and was validated on the partner Toyota vehicle.

4.2.2 Software Architecture

The modular software layers are designed according to ISO/IEC/IEEE 42010 software architecture standards [iso, g], supporting scalability, maintainability, and interoperability. Each layer's function is mapped to its implementation in Chapter 5.9:

- **Data Acquisition Layer:** Captures sensor and telemetry data, applies SMA and median filters, and performs anomaly threshold checks. Implementation details are in Chapter 5.9.
- **Processing Layer:** Performs edge anomaly detection, timestamping, and data aggregation. Standards for real-time processing and task scheduling ensure predictable latency. Implementation is in Chapter 5.9.
- **Cloud Integration Layer:** MQTT-based JSON payloads are used for cloud ingestion, following OASIS MQTT v5 and ISO/IEC 27001 security standards [iso, 2022a]. TLS/SSL ensures encrypted transmission (Chapter 5.9.
- **User Interface Layer:** Dashboards display real-time fuel levels, GPS locations, anomaly alerts, and historical trends. Web standards (HTML5, CSS3, JavaScript)

and secure REST APIs are employed, as described in Chapter 5.9.

4.2.3 Machine Learning and Algorithms

Machine learning components follow ISO/IEC 29119 software quality standards [iso, 2022b] and IEEE P7003 AI accountability guidelines [iee,]:

- **Models:** Isolation Forest was selected as the production detector because Model C (contamination = 0.015, `n_estimators` = 300, `max_samples` = 256, threshold = training-set q995 + IQR margin + 0.003) was the only configuration meeting both $\geq$90% precision and $\leq$10% FPR under leave-one-trip-out evaluation (precision = 0.966, recall = 0.966). Matrix Profile (window = 12, z-threshold = 2.0) was retained as a supporting subsequence detector for slow-leak and siphon patterns that single-point scoring is less sensitive to.
- **Evaluation Metrics:** Confusion matrices, ROC curves, and precision–recall graphs are used for model validation, ensuring alignment with ISO/IEC 25010 software quality characteristics. Chapter 5.9, Chapter 5.10 provides practical application in performance evaluation.
- **Alternative Models:** Autoencoders and LSTM networks were considered but excluded due to edge processing constraints, as justified in Chapter 4.3.

4.2.4 Communication and Security

Communication protocols and security measures are aligned with ISO/IEC 27018, ISO/IEC 27001, and GDPR standards to ensure data privacy, encryption, and secure access control:

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- **MQTT Protocol:** Supports lightweight, reliable IoT messaging, implemented with QoS 1 to guarantee delivery. Applied in Chapter 5.9.
- **TLS/SSL Encryption:** Provides end-to-end encryption for transmitted data. Compliance with ISO/IEC 27001 ensures data integrity and confidentiality. Applied in Chapter 5.9.
- **Role-Based Access and Anonymization:** Enforces secure data access policies and anonymizes sensitive driver information, complying with ISO/IEC 27018 and GDPR principles. Applied in Chapter 5.9.
- **JSON Serialization:** Standardized data format for cloud compatibility, ensuring interoperability with cloud services and dashboards. Applied in Chapter 5.9.

Wide-area connectivity uses a SIMCom A7670E LTE (Cat-1) module, chosen over a GSM-only baseline because it provides an LTE-primary, 2G-fallback hierarchy in one device. LTE is the primary path for its lowest latency and highest throughput, and because it is the network Philippine operators are expanding rather than retiring. GSM 2G on the same module is a deeper fallback for legacy-only coverage; consistent with NTC MC No. 003-09-2025 (phased 2G/3G decommissioning) it is treated as transitional, and because the A7670E is LTE-first rather than 2G/3G-only it remains compliant with that circular. LoRa provides depot, yard, and long-range telemetry; WiFi provides a depot or driver-tether path; and SPIFFS offline-buffer replay is the final fallback so no record is lost when all real-time channels are down [China Academy of Information and Communications Technology, 2025].

4.3 Alternative Software Designs and Cost Considerations

During the planning and prototype development phase, several candidate software designs were evaluated for their suitability to the system's requirements. Table 4.2 summarizes the strengths, potential risks, and fit relative to the system specifications. Matrix Profile (MP), Isolation Forest (IF), and a combined Matrix Profile + Isolation Forest (MP + IF) configuration were implemented and tested to compare their effectiveness in detecting fuel-related anomalies. The comparison considered detection accuracy, computational efficiency, and deployment feasibility on the target hardware platform. The findings from these tests were used to determine the most appropriate approach for the final system implementation.

TABLE 4.2 ALTERNATIVE SOFTWARE DESIGNS EVALUATED FOR THE PROTOTYPE

Candidate	Strengths	Risks / Trade-offs	Fit vs. Spec
Isolation Forest (IF)	Fast, unsupervised, handles multivariate data; simple to deploy using scikit-learn	Limited temporal context; sensitivity to contamination parameter may affect false alarms	High
Matrix Profile (MP)	Shape-based time-series discord detection; ideal for sudden fuel drops or irregular usage patterns; efficient in local runtime	Requires window-size tuning; primarily one-dimensional unless extended for multivariate data	Very High

A cost study was conducted after prototype testing to confirm that the system configuration was cost-effective. This study reviewed hardware, software, and development resources, comparing the chosen open-source and local execution setup against alternative configurations such as commercial analytics tools, cloud-based computing, or hardware-integrated telemetry systems.

The total estimated budget for developing and implementing the prototype system is **PHP 10,446**, confirming that the selected configuration remains the optimal and sustainable choice for achieving the project's objectives.

4.4 Summary

This chapter presented the key technical and ethical design considerations that shape the system. By grounding the project in ISO, IEEE, OASIS, GDPR, and FAIR standards, the system ensures reliable data acquisition, secure communication, validated machine learning, and responsible AI use.

These design elements establish the foundation for the procedures described in Chapter 5, bridging theoretical concepts from Chapter 3 with the practical implementation steps of the system.

Chapter 5

METHODOLOGY

This chapter presents the research methodology employed for the development of an Machine Learning-driven fuel monitoring and anomaly detection system for fleet vehicles. The methodology integrates a multi-layered approach combining sensor-level data acquisition, edge computing, IoT communication, cloud-based machine learning, and client-side visualization. The approach emphasizes real-time anomaly detection, accurate fuel monitoring, operational transparency, and actionable driver and fleet management insights.

The research methodology is structured to address both general and specific objectives, ensuring that the system is technically feasible, scalable, and reliable. Figure ?? and Table 5.1 summarize the methods and approaches for each objective.

TABLE 5.1 SUMMARY OF METHODS FOR REACHING THE OBJECTIVES

Objectives	**Methods / Approaches**	**Locations**

GO: Develop and evaluate an IoT-based system for trucks in collaboration with Hino Parañaque Subic GS Auto Inc. (SGSA) that monitors real-time fuel levels, tracks routes using GPS, and detects unusual fuel consumption using machine learning, improving driver safety, supporting rest scheduling, and facilitating proactive vehicle maintenance through automated alerts.	• Embedded MCU preprocessing and aggregation • LoRa/Cellular communication for low-latency and fail-safe data transfer • Cloud-based machine learning for anomaly detection • Web dashboard for visualization • Driver alert subsystem for operational safety	Sec. 5.9
SO1: Measure fuel levels using digital float sensors integrated via the truck's ECU/diagnostic connector and GPS module, ensuring non-invasive installation and real-time accuracy within ±5%.	• Digital fuel sensors interfaced with ECU/OBD-II • Signal filtering and smoothing • Calibration against manual measurements • Continuous validation for measurement reliability	Sec. 5.9

SO2: Transmit real-time fuel and GPS data through IoT-based communication to a cloud server, accessible through a centralized dashboard.	• Real-time IoT transmission via LoRa (primary) and Cellular Connection (backup) • Cloud ingestion APIs • Latency measurement and benchmarking to ensure timely reporting	Sec. 5.9
SO3: Capture and log GPS location data every 5 seconds via an embedded module, correlating route information with fuel and time data for behavior tracking.	• GPS-based location logging at 5-second intervals • Synchronization with fuel data • Structured cloud database for multi-source data storage and route–fuel correlation	Sec. 5.9
SO4: Design and evaluate a context-aware fuel usage deviation modeling approach using non-parametric unsupervised ML models (Matrix Profile or Isolation Forest), achieving 90% precision and 10% false positives.	• Offline training of anomaly detection models using Isolation Forest and Matrix Profile • Evaluation using precision, FPR, and accuracy • Context-aware thresholding to reduce false positives	Sec. 5.9
SO5: Develop and integrate hardware-based rest and maintenance alert mechanisms (e.g., Start/Stop button, buzzer indicators) tied to the web dashboard using cumulative fuel and distance data.	• Driver rest and maintenance alert system with cumulative distance/time tracking • Cloud-based monitoring and logging • Alert notification via dashboard and in-vehicle signals	Sec. 5.9

5.1 System Architecture and Conceptual Design

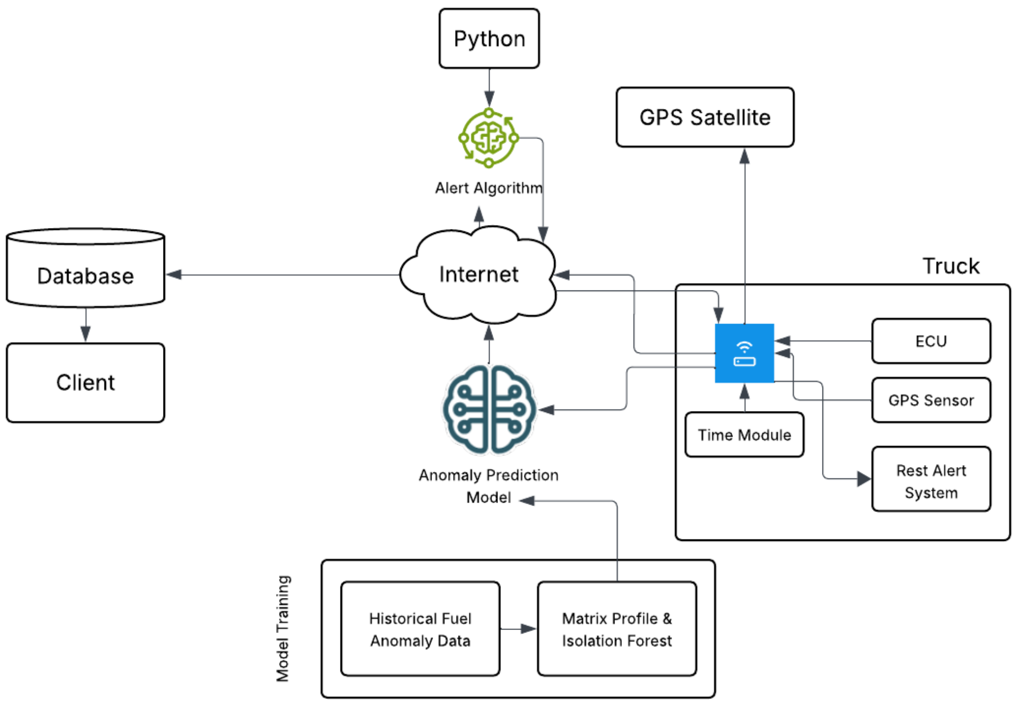

Fig. 5.1 Proposed Conceptual Diagram

The system follows a layered architecture with a focus on modularity, scalability, and reliability. The conceptual framework (Figure 5.1) integrates the following:

- **Vehicle Sensing Subsystem:** Digitally calibrated fuel-level sensors, GPS modules, and driver rest-time monitors. Sensor signals are collected via ECU or OBD-II ports.
- **Edge Processing:** The MCU performs signal preprocessing, filtering, smoothing, synchronization, and timestamping before transmission.

- **IoT Communication:** LoRa transceivers enable low-power, long-range data transmission; cellular backup, balancing cost and bandwidth considerations.
- **Cloud Intelligence:** Historical data is used to train unsupervised ML models (Isolation Forest, Matrix Profile) to detect anomalies. The cloud evaluates incoming real-time data and triggers alerts.
- **Client Visualization:** A web-based dashboard provides real-time visualization of fuel levels, routes, alerts, and historical playback for decision-making.

The system architecture (Figure 5.2) is an edge-to-cloud fleet-monitoring pipeline in which the embedded HMI performs data acquisition, filtering, trip-state handling, local driver display, and prioritised transmission. The backend stores telemetry, synchronises state across three concurrent client surfaces - the in-truck HMI, the mobile driver web application, and the administrative dashboard - and dispatches anomaly and safety alerts through the same database tables that the three surfaces subscribe to.

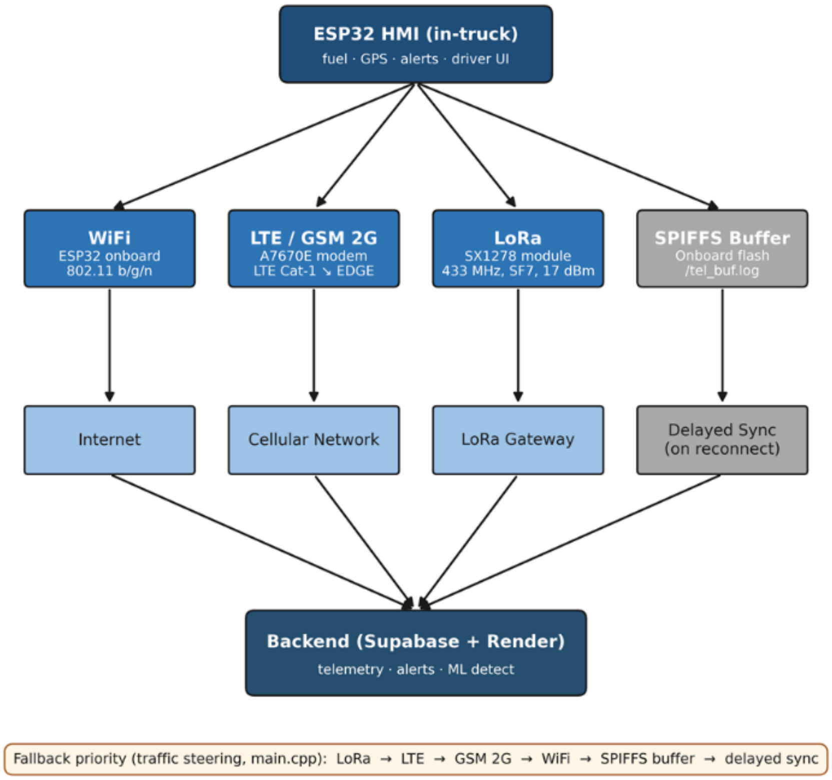

Fig. 5.2 End-to-end System Architecture

(Figure 5.2) illustrates the end-to-end communication architecture of the system, depicting how data flows from the ESP32 HMI through multiple transmission channels to the cloud backend and connected client interfaces. The HMI, serving as edge node, is responsible for collecting fuel, location (GPS), and alert data while also providing the user interface for the driver. From the HMI, data is transmitted through one of three active communication channels — WiFi (802.11 b/g/n), LTE/GSM 2G (LTE Cat-1 to EDGE), and LoRa (433 MHz, SF7, 17 dBm) — each routing through their respective network mediums of Internet, Cellular Network, and LoRa Gateway before converging at the Backend (Supabase + Render), which handles telemetry ingestion, alert dispatching, and ML anomaly detection. While the SPIFFS onboard flash buffer, which serves as the fourth passive channel, operates as a local fallback wherein the stored telemetry can be synced when a connection is re-established. The fallback priority order (LoRa $\rightarrow$ LTE $\rightarrow$ GSM 2G $\rightarrow$ WiFi $\rightarrow$ SPIFFS buffer $\rightarrow$ delayed sync) ensures continuous data delivery even under degraded network conditions. From the backend, processed data is simultaneously propagated to the mobile driver web application, the administrative dashboard, and the HMI poll-back, maintaining real-time synchronization across all three client surfaces.

5.1.1 End-to-End Telemetry Data Flow

```
START
WHILE system is running DO
    READ ECU fuel data
    READ ECU operational data
    READ GPS data
    APPLY Median Filter
```

```
APPLY EMA Filter
APPLY Dead-Band Filter
APPLY Direction-Aware Filter
// UPDATE trip state
    IF vehicle is loading THEN
        trip_state = "Loading"
    ELSE IF vehicle is moving THEN
        trip_state = "Active"
    ELSE IF temporary stop detected THEN
        trip_state = "Paused"
    ELSE IF trip completed THEN
        trip_state = "Ended"
    END IF
COMPUTE fused distance using GPS and OBD data
// SELECT available communication channel
    IF LoRa available THEN
        Send via LoRa
    ELSE IF LTE available THEN
        Send via LTE
    ELSE IF GSM available THEN
        Send via GSM
    ELSE IF WiFi available THEN
        Send via WiFi
    ELSE
```

```
            Save data to SPIFFS
        END IF
    SEND telemetry to backend
    INSERT telemetry into telemetry_logs
    RUN ML anomaly detection
    IF anomaly passes contextual gate THEN
        INSERT alert into alerts
    END IF
    UPDATE database
    UPDATE dashboard, mobile app, and HMI display
END WHILE
END
```

5.2 Hardware Methodology

TABLE 5.2 HMI GPIO MAP

Pin	**Function**
GPIO 22	`BTN_TAKE_REST` (active-low, debounced 50 ms)
GPIO 39	`BTN_SNOOZE` (active-low, debounced 50 ms)
GPIO 25	Buzzer (PWM through the LEDC peripheral)
GPIO 5	A7670E modem PWRKEY

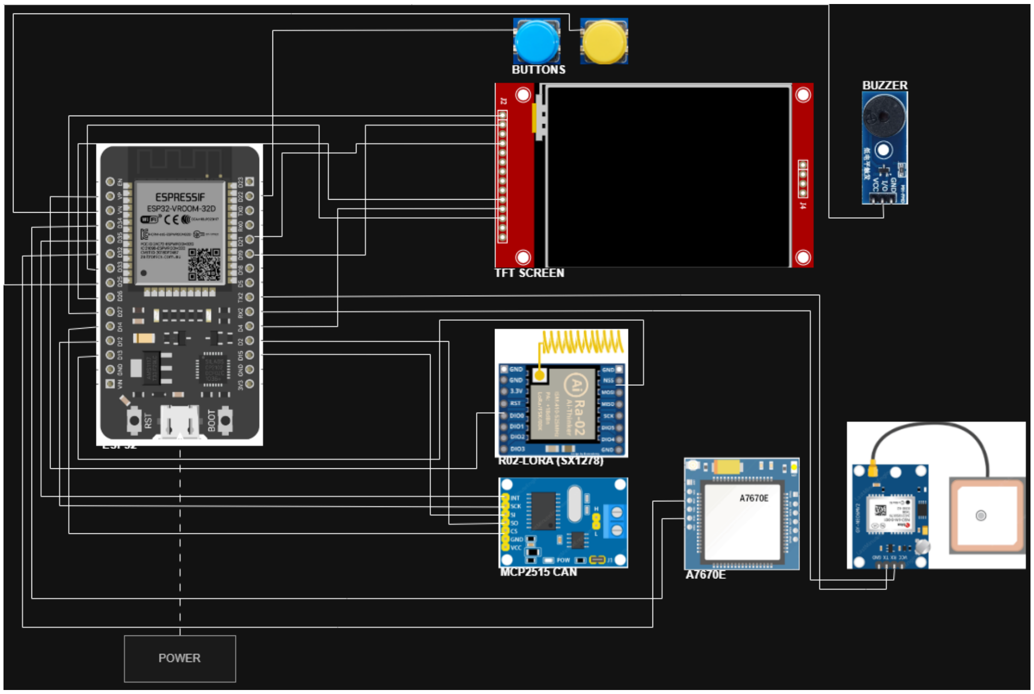

Fig. 5.3 ESP32 Hardware Block Diagram

5.3 OBD-II / ECU Fuel and Vehicle Data Acquisition Methodology

Fuel level was acquired through the Toyota meter Mode 22 PID 21/29 channel of the OBD-II interface. The same telemetry cycle also captured contextual OBD-II variables such as speed, engine RPM, engine load, throttle position, mass air flow, coolant temperature, engine runtime, and diagnostic-trouble-code status. These fields were timestamped at the device and transmitted to the backend together with GPS, trip-state, and communication-channel metadata so that the anomaly detector could interpret fuel changes in relation to vehicle operation rather than fuel percentage alone.

5.3.1 Physical Interface

The acquisition layer uses the vehicle OBD-II port and ISO 15765-4 transport over CAN. This enables the prototype to read fuel and vehicle context without tank-cap modification or sender-loom alteration.

5.3.2 Toyota Meter Mode 22 PID 21/29 Fuel Acquisition

Fuel acquisition uses Toyota meter Mode 22 PID 21/29 with a scale of 0.5 L per ADC count. The firmware polls the fuel channel on a 1.5 s cadence and applies a retry policy when the adapter returns a negative acknowledgment or missing response.

5.3.3 Co-acquired Mode 01 PIDs

TABLE 5.3 OBD-II PID TO `TELEMETRY_LOGS` COLUMN MAPPING

PID	**Quantity**	**Telemetry column**
Mode 22 0x21 0x29	Fuel sender voltage (0.5 L per ADC count)	`fuel_level` (filtered) + `fuel_source='mode22'` + `fuel_confidence`
Mode 01 0x0C	Engine RPM	`engine_rpm`
Mode 01 0x0D	Vehicle speed (km/h)	`speed`
Mode 01 0x04	Calculated engine load (%)	`engine_load_pct`
Mode 01 0x11	Throttle position (%)	`throttle_pct`
Mode 01 0x10	Mass air flow (g/s)	`maf_gps`
Mode 01 0x05	Engine coolant temperature (degC)	`coolant_temp_c`
Mode 01 0x1F	Time since engine start (s)	`engine_runtime_sec`
Mode 03 / 0x01	Stored DTCs (count + presence)	`dtc_present`, `dtc_count`

5.3.4 Validity and Confidence Fields

The firmware writes `fuel_valid`, `fuel_source`, and `fuel_confidence` on every telemetry cycle. These fields allow the backend and dashboard to distinguish valid Mode 22 fuel readings from missing, stale, or fallback values.

5.3.5 Timestamping

Each telemetry record carries the original device-side timestamp as `original_device_timestamp`, while the backend records its own receipt time. Backend reconciliation uses server time to separate transmission delay from actual sampling time.

5.3.6 Field Reference Choice

The manufacturer dashboard gauge is used as the field reference for fuel-validation activity. This choice follows the delimitations in Chapter 1, because direct manual dipstick validation and tank-cap opening were not permitted by the partner fleet operator.

5.4 Fuel Preprocessing Methodology

The raw OBD-II fuel reading was passed through a multi-stage edge filter before publication. A median window rejects single-sample spikes caused by slosh and sender chatter; an exponential moving average smooths residual noise; a dead-band publisher prevents display jitter; and a direction-aware step rule permits downward movement consistent with fuel burn while rejecting upward movement during motion unless a stationary refuel signature is observed. This filtered value is the fuel percentage displayed on the HMI and stored in `telemetry_logs`.

The filter cascade begins with the raw OBD-II Toyota Mode 22 PID 21/29 fuel value sampled every 1.5 s. Stage 1 applies a median over the 31 most recent samples, corresponding to approximately 45 s of memory. Stage 2 applies an EMA with $\alpha = 0.04$,

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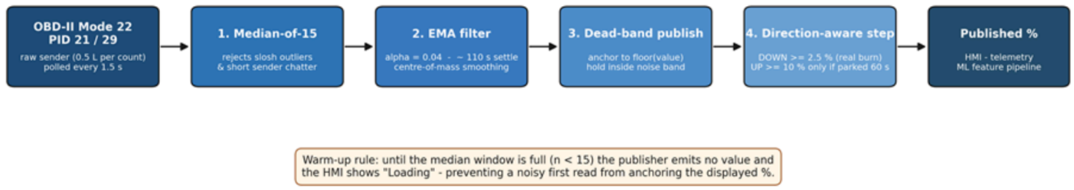

Fig. 5.4 Fuel-level filter cascade. The raw OBD-II Mode 22 PID 21/29 fuel sample (every 1.5 s) is passed through a five-stage cascade: median-of-31, EMA ($\alpha = 0.04$), dead-band publisher, direction-aware step rule, and post-refuel quiescence freeze. The published percentage feeds the HMI display and the telemetry stream.

corresponding to approximately 110 s settling. Stage 3 publishes only stable changes by anchoring the displayed percentage to the floored EMA value. Stage 4 applies a direction-aware step rule: downward movement is allowed when the EMA falls at least 2.5% below the anchor, while upward movement is allowed only when the EMA rises at least 10% above the anchor and the vehicle has been stationary for at least 60 s. Stage 5 freezes output for 5 min after a refuel event using `REFUEL_QUIESCENCE_MS`. The resulting published fuel percentage is sent to the HMI and written to `telemetry_logs.fuel_level`.

During warm-up, the first three samples per trip session are withheld from publication and the human-machine interface displays the placeholder `Loading` until the median window has been populated. This prevents a noisy first read from anchoring the displayed value at an inaccurate level.

5.5 Communication and Offline Sync Methodology

5.5.1 Priority Chain Rationale

LTE was treated as the primary wide-area data path because it offers the lowest steady-state latency and highest payload throughput among the available wide-area options. LoRa was retained as the depot/yard and long-range telemetry path because of its multi-kilometre line-of-sight reach at very low power. GSM 2G was retained as the deeper-fallback for 2G-only coverage pockets on the Batangas and Subic routes. WiFi was retained as a depot or driver-tether path. SPIFFS offline-buffer replay is the final fallback that guarantees no telemetry record is silently lost when all four real-time channels are unavailable.

TABLE 5.4 COMMUNICATION MODULE COMPARISON

Module	Technology	Hardware	Bandwidth	Range	Power	Role
WiFi	802.11 b/g/n	ESP32 built-in	approx. 10 Mbps	approx. 50 m	Low	Depot and driver hotspot
LTE	LTE Cat-1	A7670E	approx. 10 Mbps	Cell coverage	Medium	Primary cellular path
GSM 2G	EDGE	A7670E fallback	approx. 80 kbps	Wider than LTE	Medium	Cellular fallback
LoRa	SX1278, 433 MHz, SF7, BW 125 kHz	RA-02, 17 dBm TX	approx. 5.5 kbps	approx. 50 m urban NLOS at 97% measured; approx. 1 km open LOS at 60% measured; up to 10 km LOS per SX1278 datasheet	Very low	Primary telemetry path (intelligent traffic-steering chain)
Offline	-	SPIFFS flash	-	-	None	Outage replay

The LoRa range column is reported as two values: the operational urban NLOS reach measured in this study and the line-of-sight reach published in the SX1278 datasheet. LoRa is the highest-priority transport in the intelligent-traffic-steering chain (LoRa → LTE → GSM 2G → WiFi → SPIFFS buffer); the cellular path provides coverage of the urban-NLOS regime past approximately 50 m where LoRa delivery degrades.

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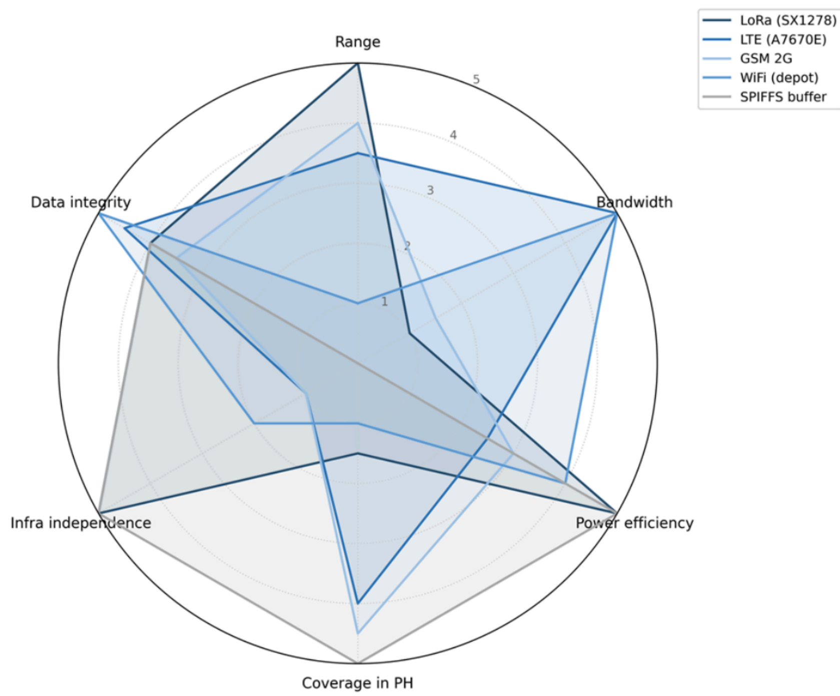

Fig. 5.5 Communication Method Comparison radar. Five channels (LoRa, LTE, GSM 2G, WiFi, SPIFFS) are compared on six axes: range, bandwidth, power efficiency, Philippine coverage, infrastructure independence, and data integrity, normalised to a five-point scale.

5.5.2 Telemetry JSON Schema

```
{
  "truck_id": "<uuid>",
  "driver_id": "<uuid>",
  "trip_id": "<uuid>",
  "sent_at": "<iso-8601>",
  "original_device_timestamp": "<iso-8601>",
  "source_device": "hmi" | "mobile",
  "fuel_level": <0..100>,
  "fuel_valid": <bool>,
```

```
"fuel_source": "...",
"fuel_confidence": <0..1>,
"lat": <float>,
"lon": <float>,
"gps_source":
  "embedded_gps" | "mobile_gps" | "fused_mobile_fallback",
"gps_accuracy_m": <float>,
"gps_heading_deg": <float>,
"mobile_speed_mps": <float>,
"speed": <km/h>,
"odometer_km": <float>,
"engine_status": "on" | "off" | "idle",
"seq": <bigint>,
"event_id": "<uuid>",
"channel_used":
  "lora" | "gsm" | "offline_replay" | "mobile_app" | "simulation",
"comm_channel": "...",
"hmi_driving_sec": <int>,
"hmi_rest_sec": <int>,
"hmi_trip_status": "...",
"engine_rpm": <int>,
"engine_load_pct": <float>,
"throttle_pct": <float>,
"maf_gps": <float>,
```

```
  "coolant_temp_c": <float>,
  "engine_runtime_sec": <int>,
  "dtc_present": <bool>,
  "dtc_count": <int>,
  "distance_gps_delta_km": <float>,
  "distance_obd_delta_km": <float>,
  "distance_fused_delta_km": <float>
}
```

5.5.3 Fall-through Policy

Each telemetry record is attempted on the highest-priority available channel: LoRa first if a gateway uplink is in range, otherwise LTE, then GSM 2G, then WiFi. If an attempt fails because of a non-2xx response from the backend or a modem error, the firmware falls through to the next channel without duplicating the `seq` or `event_id`. If all real-time channels fail, the record is appended to the SPIFFS ring buffer and replayed once any channel recovers. The replayed record is tagged `channel_used = 'offline_replay'` so the backend can distinguish live from replayed data. HMAC signing of the payload and HTTPS/TLS transport protect telemetry against tampering and eavesdropping on the cellular path.

5.5.4 LoRa Range Test Protocol

The LoRa physical-layer range test was conducted with the SX1278 module configured at seventeen-dBm transmit power, four-hundred-thirty-three-megahertz centre frequency,

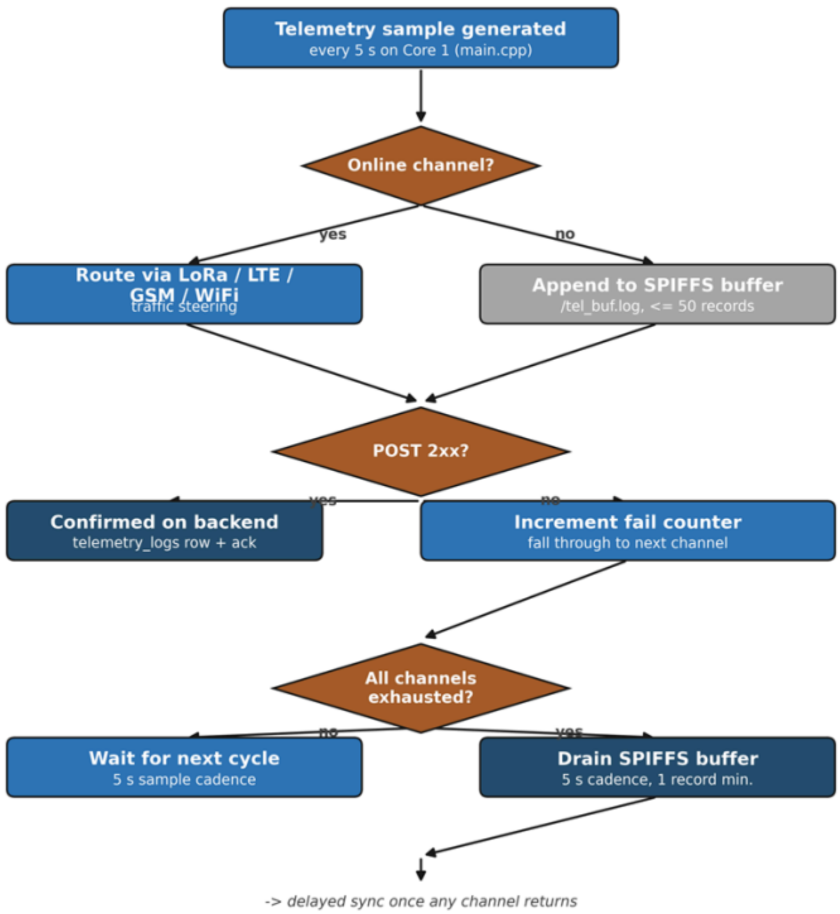

Fig. 5.6 Offline-buffer fall-through flowchart. The decision tree executed by the Core 0 worker each five-second sample cycle first routes telemetry through the highest-priority online channel. On a non-2xx response, the worker falls through to the next channel; if all channels are exhausted, the record is appended to `/tel_buf.log`. The buffer drains on a five-second cadence with a guaranteed-forward-progress minimum of one record per worker pass.

spreading factor seven, and one-hundred-twenty-five-kilohertz bandwidth. One hundred telemetry-sized packets were transmitted from the in-truck node to a fixed gateway at each of the open line-of-sight distances of ten, fifty, one hundred, two hundred and fifty, five hundred, one thousand, two thousand, and five thousand metres, and at the urban non-line-of-sight distances of one hundred, two hundred and fifty, and five hundred metres. The received-signal-strength indicator, the signal-to-noise ratio, and the successfully-received-packet count were recorded at the gateway for each distance band.

5.5.5 GSM / LTE Drive-test Protocol

Carrier-reported signal quality was logged via the AT+CSQ command issued to the A7670E modem at five locations of operational interest: the depot, a Metro Manila urban environment, a highway corridor, a tunnel or underpass selected to trigger the buffer-activation path, and a provincial route in Batangas exhibiting two-gigahertz-only coverage. The per-packet upload latency was computed at the backend as the difference between the device-side `sent_at` timestamp and the backend-side receipt timestamp; the fiftieth- and ninety-fifth-percentile latencies are reported per location alongside the successful-upload fraction.

5.5.6 Packet-delivery and Buffering Protocol

A drive route was executed that traversed three distinct connectivity regions: an urban region with strong LTE coverage, a highway region with marginal LTE coverage, and a confirmed no-signal dead zone. One hundred packets were generated at each stage; the per-stage outcome (sent on first attempt, sent after a retry, buffered to the on-device

SPIFFS log, replayed once connectivity returned, or lost) was tabulated from the firmware serial-output log and the backend-side telemetry receipt log.

5.6 Backend and Database Methodology

5.6.1 Backend Stack

The backend stack consists of Node/Express hosted on Render and Supabase Postgres as the managed relational database. This stack receives device telemetry, manages trip state, runs alert rules, invokes the anomaly-detection service, and exposes synchronized state to the HMI, mobile driver application, and administrative dashboard.

TABLE 5.5 SUPABASE POSTGRES SCHEMA

Table	Purpose	Key columns
`users`	Accounts (head_admin / fleet_manager / manager / driver)	`id`, `role`, `assigned_truck_id`
`trucks`	Vehicle profile + maintenance state	`id`, `truck_code`, `plate_number`, `current_odometer_km`, `status`
`trip_sessions`	Active / paused / ended trips with destination, route polyline, nav state, mobile-GPS columns	`id`, `truck_id`, `driver_id`, `trip_status`, `distance_km`, `assigned_destination`, `route_steps`, `mobile_lat/lon`
`telemetry_logs`	Per-cycle fuel + GPS + OBD context + anomaly flag	`fuel_level`, `lat/lon`, `speed`, `odometer_km`, `channel_used`, `anomaly_flag`, `anomaly_score`, `model_source`
`archived_telemetry_logs`	Older trip telemetry beyond retention window	same as `telemetry_logs` + `archived_at` + `archive_reason`
`trip_summaries`	Per-trip aggregates upserted by `refresh_trip_summary()` RPC	`total_distance_km`, `total_alerts`, `total_anomalies`, `average_fuel_level`
`alerts`	Rest / maintenance / overspeed / low_fuel / fuel_anomaly events	`alert_type`, `severity`, `is_resolved`, `trip_id`
`latency_logs`	SO2-c latency tracking (sent_at vs received_at)	`latency_ms`, `channel_used`, `retry`, `sent_at`, `received_at`

Figure 5.7 presents the Entity-Relationship Diagram (ERD) of the system's Supabase

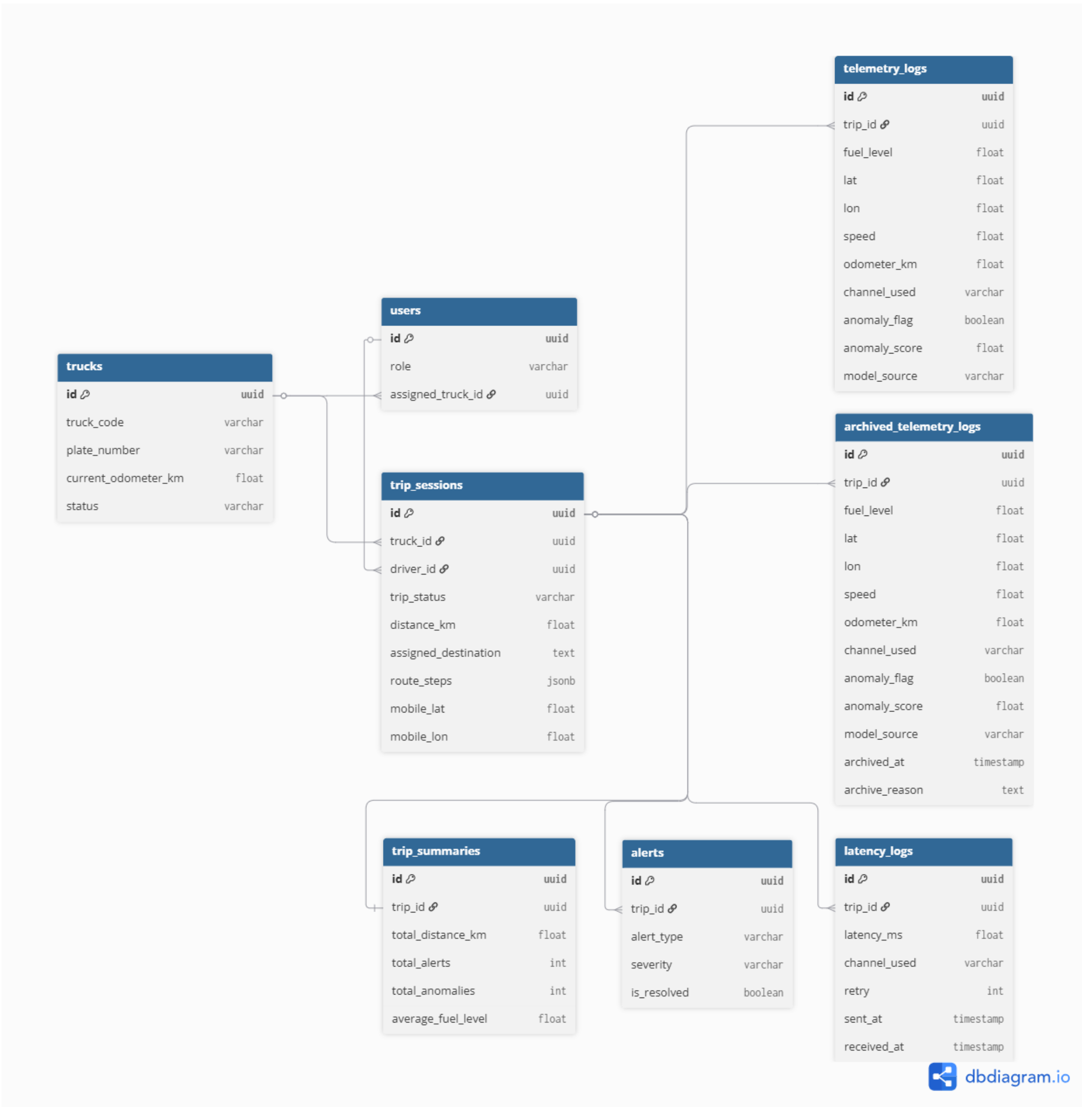

Fig. 5.7 ERD Diagram

PostgreSQL database, containing eight interrelated tables: `trucks`, `users`, `trip_sessions`, `telemetry_logs`, `archived_telemetry_logs`, `trip_summaries`, `alerts`, and `latency_logs`. The `trip_sessions` table serves as the main hub for the relationships among these tables, linking trucks (`truck_id`) and drivers (`driver_id`). Other tables such as `telemetry_logs`, `archived_telemetry_logs`, `trip_summaries`, `alerts`, and `latency_logs` reference trips by `trip_id` to relate records to a specific trip session. This data structure is designed to maintain referential integrity throughout the schema and support scalable access to the HMI dashboard, mobile app, and backend analytic pipeline.

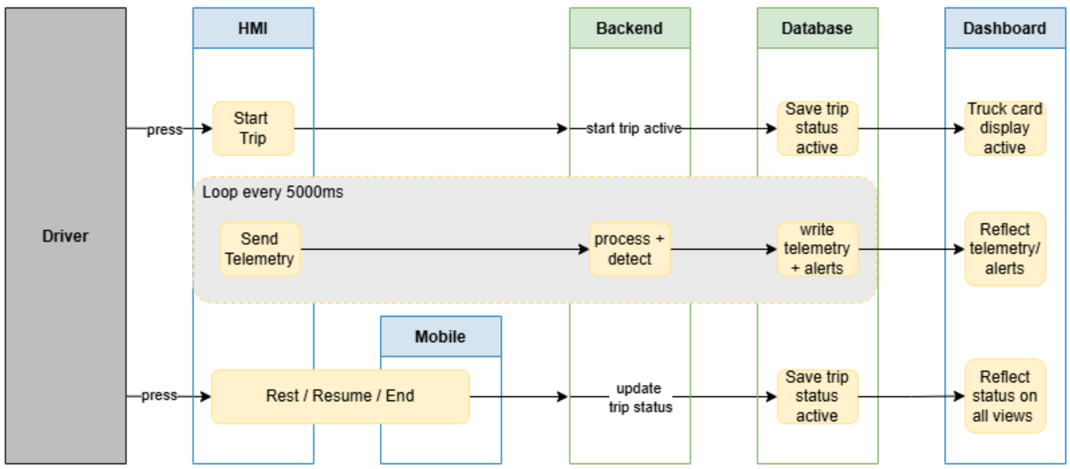

Fig. 5.8 Cross-Interface State-Synchronization Sequence Diagram

Figure 5.8 illustrates the cross-interface synchronization flow of the system, depicting how data propagates across the HMI, backend, database, and dashboard layers in response to driver actions. When the driver initiates a trip via the HMI or mobile app, the backend creates an active trip record in `trip_sessions`, which simultaneously triggers the truck card display to activate on the dashboard. Telemetry data is transmitted every 5000 ms as a cycle during a trip. Each cycle is processed and anomaly-detected by the backend before it is recorded in telemetry_logs and displayed as alerts in the dashboard. Trip status changes are also reflected throughout the system through trip_sessions.trip_status for Rest, Resume, or End trip states. This mechanism ensures that all interfaces, including the mobile PWA and HMI poll-back, remain synchronized with the current session state. Throughout this process, trip_sessions functions as the canonical source of trip data and the authoritative state controller for the entire architecture.

5.6.2 Odometer and Trip-distance Methodology

The Toyota Hilux vehicles used in the prototype display lifetime mileage on the physical dashboard, but that value is not exposed through the OBD-II adapter available to the system. The firmware probes the Toyota-specific Mode 21 and Mode 22 odometer data identifiers, but the fleet does not return a usable lifetime odometer value; as a result, telemetry_logs.odometer_km is either null or a trip-scoped speed-integrated fallback generated by the HMI. It is not treated as an absolute lifetime odometer reading.

For this reason, backend lifetime mileage is computed from completed trip distance rather than from the raw OBD odometer field. Each active trip continually updates trip_sessions.distance_km through public.compute_trip_distance_km(p_trip_id), implemented in backend/db/migration_distance_rpc_v3_jitter.sql. The function takes the greater of capped speed-by-time integration and filtered GPS haversine distance, rejecting stationary GPS jitter below the movement threshold. When a trip transitions to trip_status = 'ended', backend/server.js calls public.accumulate_trip_distance_odometer(which adds that finalised distance_km to trucks.current_odometer_km exactly once and records trip_sessions.odometer_accumulated_at.

TABLE 5.6 ALERT RULE THRESHOLDS IMPLEMENTED

Alert	Threshold	Trigger source	Hardware action
Rest reminder	distance $\geq$ 300 km AND drive time $\geq$ 4 h	backend `/device/alerts`	HMI buzzer, screen alert, `Take Rest` button
Maintenance (oil)	trip distance $\geq$ 5,000 km	backend trip aggregation	Dashboard banner + planned email/push
Overspeed	speed $\geq$ 100 km/h	telemetry stream	HMI flash and log
Fuel anomaly	ML flag from Isolation Forest or Matrix Profile	`anomaly_service.py`	Dashboard alert card

TABLE 5.7 EXTENDED SAFETY IMPLEMENTATION STATUS

Safety alert	Status	Implementation note
Overspeed	Implemented	`backend/server.js:758-772`, threshold 100 km/h
Rest reminder	Implemented	SO5-b spec is 300 km / 6 h; implemented at 4 h per LTO driver-fatigue reading
Maintenance (5,000 km)	Implemented	`backend/server.js:879-894`
Harsh acceleration / braking	Partial	`acceleration_kmph_per_sec` computed at `ml/preprocess.py:74`; no alert rule yet
Lane departure	Out of scope	Requires camera and IMU - future work
Fatigue / drowsiness	Out of scope	Requires camera with face detection - future work

5.7 Machine Learning Methodology

The anomaly-detection pipeline uses Isolation Forest Model C as the primary production detector and Matrix Profile as a supporting subsequence detector. Model C receives context-aware features derived from fuel level, speed, distance, engine operation, driver behavior, route context, and fuel-economy baseline deviation. Matrix Profile is retained to support detection of gradual subsequence-level fuel-loss patterns. Model performance is evaluated using leave-one-trip-out validation so each held-out trip is scored by models trained and calibrated on separate trips.

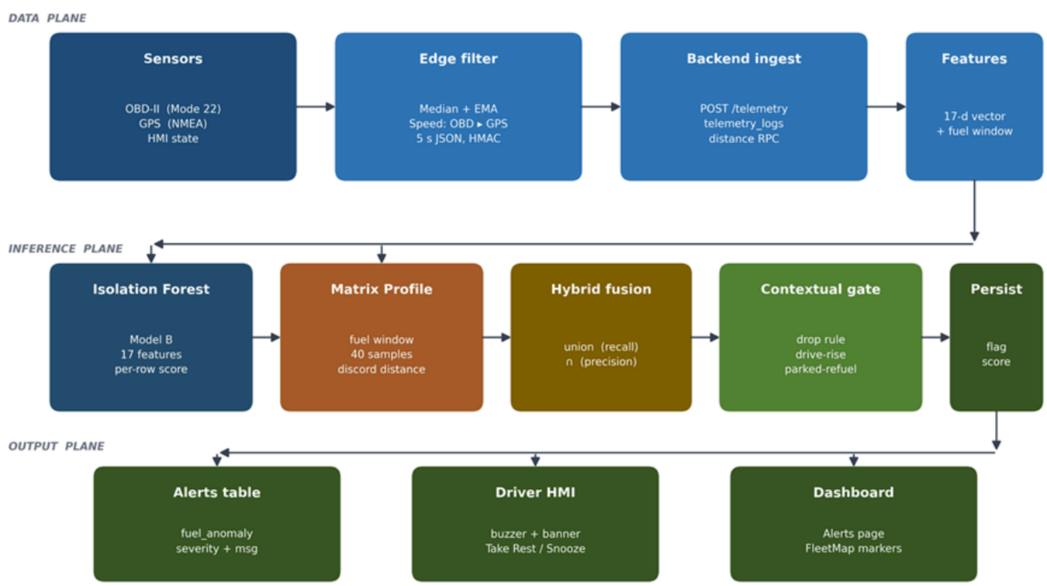

Fig. 5.9 ML Anomaly Detection Architecture. The pipeline includes sensor sources, edge preprocessing, backend ingest, Supabase storage, 33-feature Model C vectors, dual-model inference, hybrid fusion, and downstream consumers including persisted anomaly flags, alerts, HMI buzzer and banner, and the React dashboard.

TABLE 5.8 DATASET COMPOSITION FOR EVALUATION CORPUS

Dimension	**Value**
Total telemetry rows	20,412
Total trips	79
Anomalous trips	6
Normal samples (clean)	20,327
Anomalous samples (labelled)	85
Source partition	`public_geolife_route_derived_fuel`: 19,778 rows; `project_native`: 634 rows
Label class	`clean_public_route`: 19,778 rows; clean (project-native): 465 rows; `controlled_injection`: 169 rows
Evaluation subset	`public_clean` (out-of-distribution): 19,778 rows; `project_native_clean` (in-distribution baseline): 465 rows; `project_native_injected` (headline target): 169 rows
Trip length distribution	Short trips (<12 min): 30; medium trips (12–30 min): 31; long trips (>30 min): 18

5.7.1 Anomaly Injection Methodology

Controlled-injection anomalies were applied to the project-native partition by `ml/inject_anomalies.py` across three industry-documented patterns: `sudden_fuel_drop` (single-step drop of at least 10%), `abnormal_rapid_decrease` (4-sample drop window), and `gradual_leak` (slow steady decrease at less than 1% per step). Of the 85 labelled rows, 76 are `gradual_leak` by design, because gradual leaks are the hardest detection regime: each row sits inside the sensor-noise envelope, and the injection budget deliberately weights toward that class.

TABLE 5.9 ANOMALY SCENARIO COVERAGE

Scenario	**Description**	**Primary detector**	**Coverage**
Sudden fuel theft	Single-step drop of at least 10%	Isolation Forest	Yes
Gradual siphoning	Slow steady decrease	Matrix Profile	Yes
Continuous leak	Linear decline at speed = 0	Matrix Profile	Yes
Sensor spike	Brief out-of-range raw value	Median filter (pre-ML)	Yes
Heavy legitimate consumption	Loaded uphill drive	Both models correctly classify as normal	Yes

TABLE 5.10 FEATURE SET ACROSS ISOLATION FOREST VARIANTS (A / B / C)

#	**Feature**	**Source signal**	**What it captures**	**A**	**B**	**C**
1	`fuel_level_filtered`	OBD-II PID 21/29	Baseline tank level after filter cascade	Y	Y	Y
2	`speed_filtered`	OBD-II PID 0x0D + GPS fallback	Smoothed vehicle speed	Y	Y	Y
3	`fuel_delta`	`fuel_level_filtered`	Per-row consumption increment	Y	Y	Y
4	`odometer_delta`	OBD-II odometer / integrated speed	Distance moved per sample	Y	Y	Y
5	`fuel_per_km`	`fuel_delta` / `odometer_delta`	Efficiency rate - siphoning under load	Y	Y	Y
6	`fuel_rate_per_hour`	`fuel_delta` / dt	Continuous-leak signature	Y	Y	Y

#	Feature	Source signal	What it captures	A	B	C
7	`distance_per_fuel`	1 / `fuel_per_km`	Light-load / coasting regime	Y	Y	Y
8	`route_segment_performance`	Per-segment mean vs trip mean	Route-specific consumption shift	Y	Y	Y
9	`driver_behavior_score`	Aggregate of accel + variability + idle	Long-tail driving-style signal	Y	Y	Y
10	`engine_rpm_normalized`	OBD-II PID 0x0C	Engine work intensity	Y	Y	Y
11	`engine_load_pct_norm`	OBD-II PID 0x04	ECU-reported load fraction	Y	Y	Y
12	`throttle_pct_norm`	OBD-II PID 0x11	Driver demand	Y	Y	Y
13	`fuel_drop_while_stationary`	`fuel_delta` + speed	Siphoning signature - drop while parked	Y	Y	Y
14	`fuel_drop_with_engine_off`	`fuel_delta` + `engine_status`	Theft-at-rest signature	Y	Y	Y
15	`idle_duration_score`	Rolling 5-row idle flag sum	Sustained-idle context for parked-refuel logic	Y	Y	Y
16	`fuel_drop_per_rpm`	`fuel_per_km` / `rpm_norm`	Disagreement: fuel use vs engine work	Y	Y	Y
17	`fuel_drop_per_load`	`fuel_per_km` / `engine_load_norm`	Disagreement: fuel use vs reported load	Y	Y	Y
18	`motion_state_code`	speed + `engine_status`	Motion-regime category for slosh suppression	Y	Y	Y
19	`time_since_refuel_norm`	Refuel-event detector	Refuel-window exemption signal	Y	Y	Y
20	`post_refuel_flag`	Refuel-event detector	Hard-coded 60-sample refuel exemption	Y	Y	Y
21	`delta_engine_load_norm`	`engine_load_pct_norm` delta	Rate of change of load	Y	Y	Y
22	`delta_throttle_norm`	`throttle_pct_norm` delta	Rate of change of driver input	Y	Y	Y
23	`load_per_speed_norm`	load / speed	Load relative to motion - high load while parked	Y	Y	Y
24	`expected_consumption_residual`	Physics model vs observed	Slosh-suppression baseline	Y	Y	Y

#	Feature	Source signal	What it captures	A	B	C
25	`rpm_high_share`	`engine_rpm_normalized`	Rolling fraction with RPM > 2000 (Barbado high band)	-	Y	Y
26	`rpm_red_share`	`engine_rpm_normalized`	Rolling fraction with RPM > 3500 (red band)	-	Y	Y
27	`rpm_orange_share`	`engine_rpm_normalized`	Rolling fraction 2500–3500 (orange band)	-	Y	Y
28	`rpm_yellow_share`	`engine_rpm_normalized`	Rolling fraction 1500–2500 (yellow band)	-	Y	Y
29	`speed_over_90_share`	`speed_filtered`	Rolling fraction with speed > 90 km/h	-	Y	Y
30	`speed_over_120_share`	`speed_filtered`	Rolling fraction with speed > 120 km/h	-	Y	Y
31	`kmpl_deviation_pct`	Per-asset fuel-economy baseline	Per-asset fuel-economy deviation	-	-	Y
32	`driver_kmpl_deviation_pct`	Per-driver fuel-economy baseline	Per-driver fuel-economy deviation	-	-	Y
33	`route_type_code`	Trip metadata + GPS context	Route-context disambiguator for KMPL features	-	-	Y

The feature set is based on `ml/preprocess.py` (`CONTEXT_FEATURE_COLUMNS_V2`, `DRIVING_ZONE_FEATURE_COLUMNS`, and `KMPL_DEVIATION_FEATURE_COLUMNS`), `embedded/hmi_phase1/src/main.cpp:4286-4445` for the telemetry schema, and `ml/anomaly_service.py:128` for the inference endpoint. Model A contains 24 context-aware baseline features. Model B contains 30 features by adding Barbado driving-zone features for ablation only. Model C contains 33 features by adding Habib KMPL baseline-deviation features and is the production configuration.

5.7.2 Isolation Forest Training Methodology

The Isolation Forest is trained with contamination = 0.015, because theft is genuinely rare in fleet operation, with an expected prior around 1–2%; `n_estimators` = 300; and `max_samples` = 256. The score threshold for each leave-one-trip-out fold is the 0.985 quantile of the fold's clean training scores. The choice is internally consistent with the contamination = 0.015 hyperparameter, because both expect the same 1.5% nominal anomaly fraction at the boundary, and avoids any hand-tuned magic number. No precision margin and no per-context shimming are applied.

5.7.3 Matrix Profile Methodology

Matrix Profile v2 is implemented in `ml/matrix_profile.py` with the post-discord magnitude gate ($\geq$1% absolute fuel change), direction gate (downward-only), and refuel-exemption gate calibrated against the held-out clean project-native trips at each LOTO fold. The narrow discord window ($m = 4$ samples) is intentional: it preserves subsequence sensitivity for slow-leak signatures and explains why the strict row-level IF AND MP intersection collapses to roughly 5% recall and is rejected as a deployment definition.

5.7.4 Hybrid Fusion Methodology

Two fusion strategies are characterised in `ml/infer.py::run_combined_inference()`. The Hybrid Union (IF OR MP) maximises recall. The Hybrid Intersection used for production alerting is a trip-level MP gate over row-level IF predictions: a row is flagged if and only if (a) the Isolation Forest flagged that row and (b) the trip's overall Matrix Profile decision was "anomalous shape detected". The trip-gated definition preserves the precision intent of the intersection while avoiding the granularity-mismatch penalty of strict row-level AND.

5.7.5 Contextual Gate Methodology

The contextual gate at `backend/server.js:3048-3063` receives every row flagged by the ML layer and applies three deterministic rules: drop (suppress repeated flags inside the same window), drive-rise (suppress flags during legitimate post-acceleration recovery), and parked-refuel (suppress flags during the 5-minute post-refuel quiescence window). Across the 37.7-day live deployment, the gate rejected 75.1% of model flags, reducing the user-facing alert rate from 2.90% to 0.72%.

5.7.6 Live Deployment Integration

The ML service is a Flask microservice at `ml/anomaly_service.py` exposing `POST /detect`. After each `/telemetry` insert, the backend invokes `/detect` via `setImmediate`; the result is written back to `telemetry_logs.anomaly_flag`, `anomaly_score`, and `model_source`, and a row is appended to the `alerts` table with `alert_type = 'fuel_anomaly'` when the contextual gate permits.

5.8 Evaluation Methodology

All Chapter 6 SO4 numbers are produced by a leave-one-trip-out (LOTO) evaluation that holds the calibration set strictly disjoint from the test set. For each of the twelve project-native trips, the three Isolation Forest variants and the Matrix Profile detector are refit and recalibrated on the clean rows of the remaining eleven project-native trips only; the held-out trip is then scored with that fold's models. Each prediction is therefore strictly out-of-sample with respect to both model fitting and threshold calibration. The Isolation Forest threshold for every variant is the 0.985 quantile of the fold's clean training scores, internally consistent with the contamination = 0.015 hyperparameter; no precision margin and no per-context shimming are applied.

TABLE 5.11 OBJECTIVE, PROTOCOL, AND METRIC

Objective	Evaluation protocol	Metric
SO1 - Fuel accuracy	Dashboard-gauge walkaround pairs + per-tank-band drift/noise/jitter on $n = 20,363$ valid samples	MAPE, max error, mean absolute noise (%), drift at speed = 0
SO2 - Communication	LoRa distance test (10 m to 5 km), GSM/LTE drive test at five locations, three-region packet-delivery drive (urban/highway/dead-zone)	Median + p95 latency, packet-success fraction, SPIFFS-replay share, RSSI/SNR per band
SO3 - GPS	`sent_at` interval analysis on live telemetry; GPS fix availability; route vs OSRM polyline comparison	Median + p95 inter-sample dt, fix availability (%), per-segment fuel (% per km)
SO4 - ML	LOTO cross-validation on `project_native_injected` + threshold/IQR baseline + Hybrid Union/Intersection ablation + OOD stress test on `public_clean` + 37.7-day live deployment audit	Precision, recall, FPR, F1, Wilson 95% CI, per-model confusion matrices, live alert-resolution rate
SO5 - Alerts	Alert-trigger verification across controlled-environment trip set	Trigger correctness (TP rate), driver-response capture, HMI/dashboard/mobile sync correctness

5.9 Implementation

5.9.1 Data Acquisition and Preprocessing

Data acquisition and preprocessing are crucial for reliable anomaly detection. The methodology includes:

1. **Sensor Calibration:** Fuel-level sensors are calibrated using known volumes, verified with manual measurements, and cross-checked with historical data trends.

2. **Signal Filtering:** Raw sensor signals are filtered using moving-average and median filters to reduce noise caused by vehicle motion or fuel sloshing.

3. **Synchronization:** Fuel, GPS, and operating-hour data are timestamped and synchronized to ensure consistent data alignment for model evaluation.

4. **Feature Engineering:** Derived features include fuel consumption rate, distance-to-fuel ratio, route segment performance, and driver behavior metrics, which feed into anomaly detection models.

Study Procedures

Upon voluntary agreement to participate, the study procedures will be conducted as follows:

1. Before the trip begins, a member of the research team will install a monitoring device or IoT module unit in the truck assigned to the participant.

2. Once installed, the device will automatically record fuel-level readings at regular intervals, distance and travel time, and GPS location data during normal vehicle operations.

3. For randomly selected trips, participants may be asked to perform a supervised task simulating fuel draining from the tank. This procedure will be conducted to test the accuracy of the prototype's anomaly detection function.

4. Participants will not be required to change their normal driving behavior during the data collection period.

5. After the data collection period is completed, a member of the research team will remove the monitoring device from the vehicle.

6. The study will also conduct a survey with the fleet administrator to gather feedback on the device's functionality and any problems encountered during its use. The survey will not be administered to the drivers and will not include personal questions directed at them.

Risks and Discomforts

This study involves minimal risk. The following potential risks and discomforts have been identified:

- The monitoring device will not control, modify, or interfere with the vehicle's systems and will not affect driving performance or safety. It will passively record only fuel-level data, GPS data, and travel information.
- Participants may experience minor inconvenience or brief disruption during the installation and removal of the monitoring device before and after the trip.
- GPS location data will be collected during normal operations, which may raise privacy concerns. However, the data will be anonymized through a unique code number assigned to each dataset and will be shared only with authorized members of the research team and Hino Paranaque's fleet administrator for academic research purposes.
- No foreseeable psychological, physical, or occupational risks are expected to result from participation in this study.

Steps to Minimize Risks

The research team will undertake the following measures to minimize and mitigate potential risks:

- The monitoring device will be installed and removed only by a member of the research team to ensure that the vehicle's systems are not disrupted.
- All collected data will be anonymized by replacing participant names with unique code numbers before analysis.
- Access to raw data will be restricted to authorized members of the research team and Hino Paranaque's fleet administrator only.
- All digital data will be stored in a secure, password-protected system to prevent unauthorized access.
- Physical documents, such as signed consent forms, will be kept in a locked cabinet accessible only to the principal investigator and faculty adviser.
- Data will be used only for academic purposes, and only summarized and anonymized results will be included in reports or publications.

Benefits

Participants will not receive direct financial benefits from participation. However, the results of this study may contribute to the following non-monetary benefits:

- **For the partner company and fleet managers:** The system may improve fuel efficiency monitoring and support the detection and prevention of fuel misuse, leakage, or theft.
- **For the transportation and logistics industry:** The study contributes to research on non-parametric machine learning approaches for fleet management, which may inform future industry practices.
- **For the public and the environment:** Improved fuel monitoring systems may help reduce fuel waste and contribute to lower carbon emissions.
- **For the research community:** The findings may add to academic knowledge on IoT-based monitoring systems and anomaly detection methods in real-world applications.

Compensation

Participants will not receive any compensation, tokens, or gifts for their participation in this study.

Data Storage, Duration, and Disposal

Storage of Digital Data: All digital data collected by the monitoring device will be stored in a Supabase cloud database, a secure and password-protected platform accessible only to authorized members of the research team. Files and database entries will be protected by unique credentials known only to the principal investigator and faculty adviser.

Storage of Physical Data: Signed consent forms and any printed documents will be stored in a locked filing cabinet located at the research team's office. Only the principal investigator and faculty adviser will have access to these physical documents.

Duration of Storage: All digital and physical data will be retained for five (5) years from the date of completion of the study, in accordance with De La Salle University's data retention policy.

Disposal: After the retention period, all digital data will be permanently deleted from the Supabase database and any local copies using secure deletion procedures. All physical documents will be cross-shredded and disposed of in a secure manner.

Confidentiality

For Digital Data: Participant names will not appear in any reports, datasets, publications, or presentations. A unique code number will be assigned to each participant's data in place of the participant's name. All digital files will be stored in a password-protected Supabase cloud database accessible only to the research team. Data will be used strictly for academic research purposes, and only summarized and anonymized results will be presented in the final thesis or any related publications.

For Physical Data: Signed consent forms and any printed documents will be stored in a locked cabinet. Only the principal investigator and faculty adviser will have access to these hard copy documents.

Withdrawal Procedure

Participation in this study is entirely voluntary, and participants may withdraw at any time without penalty. To withdraw, participants may follow these steps:

1. Inform the principal investigator, Samuelle P. Barja, of the decision to withdraw, either verbally or in writing.

2. Contact the principal investigator through the following:
   - samuelle_barja@dlsu.edu.ph
   - 09564054471
3. Upon notification of withdrawal, the research team will immediately stop collecting data from the participant's assigned vehicle.
4. Any data already collected that is identifiable to the participant will be permanently deleted from all storage systems within a reasonable timeframe.
5. Participants will not be required to provide any reason for withdrawal, and the decision to withdraw will not result in any consequence to their employment or relationship with their employer.

5.9.2 System Methodology

The system was implemented as follows:

- **MCU and Edge Layer:** ESP32/NodeMCU microcontrollers for filtering, timestamping, and initial anomaly scoring.
- **Vehicle Interfaces:** CAN bus or OBD-II for fuel-level readings; GPS module for location and speed tracking.
- **Communication Layer:** LoRa transceivers for real-time cloud updates; cellular backup for critical alerts.
- **Cloud Layer:** Python-based ML models for anomaly detection deployed on cloud servers; REST APIs for data ingestion; PostgreSQL database for structured storage.

- **Visualization Layer:** Web dashboard for real-time monitoring, historical trend analysis, and alert management.

5.9.3 Anomaly Detection and ML Methodology

The anomaly detection workflow is as follows:

1. **Offline Model Training:** Historical datasets are used to train Isolation Forest and Matrix Profile models to detect abnormal fuel usage patterns.
2. **Context-Aware Thresholding:** Thresholds are dynamically adjusted based on vehicle type, route segment, and driver behavior to reduce false positives.
3. **Evaluation Metrics:** Precision, FPR, accuracy, and detection latency are calculated; confusion matrices, ROC, and PR curves visualize performance.

5.9.4 IPO Chart and Design Pipeline

The data design outlines a cohesive pipeline which integrates sensor acquisition, synchronization, cloud transmission, and model-based analysis to enable real-time anomaly detection in fleet fuel systems. Data collection begins with embedded sensors capturing raw signals such as fuel level, fuel flow, GPS coordinates, and ignition events. These signals are preprocessed using filters and signal conditioning circuits to reduce noise and ensure clean input for analysis.

To ensure temporal alignment, all sensor outputs are timestamped and resampled onto a uniform time grid using linear interpolation. This synchronization step is critical for comparing trends across sensor modalities and detecting correlated anomalies. The processed signals are then uploaded to a cloud-based platform for storage and remote access.

For anomaly detection, the synchronized time-series data are used to train and evaluate unsupervised models such as Isolation Forest and Matrix Profile. These models are tested on a dataset composed of real-world and simulated scenarios to reflect both normal and anomalous behavior. Key performance metrics (including MAPE, timestamp drift, detection latency, precision, and false positive rate) are used to assess system effectiveness. This pipeline supports scalable, automated monitoring while laying the groundwork for further model refinement in future stages.

5.9.5 IPO Chart

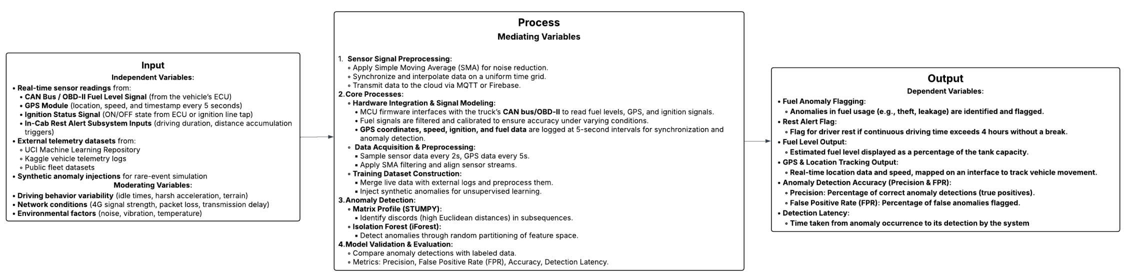

Fig. 5.10 IPO Chart

The IPO (Input-Process-Output) chart provides a structured overview of the IoT-based fleet fuel monitoring and anomaly detection system (Figure 5.10). The input data consists of real-time readings from the truck's CAN bus/OBD-II fuel signal, GPS module capturing coordinates and speed at 5-second intervals, ignition status, and in-cab rest alert subsystem signals. These inputs are supplemented by historical fuel anomaly datasets and synthetic anomalies to simulate rare or edge-case events. Moderating variables such as driving behavior (idle times, acceleration patterns, terrain), network conditions (signal strength, transmission delays), and environmental factors (temperature, vibration) also influence the data.

The process begins with signal preprocessing, where the MCU filters, calibrates, and timestamps fuel, GPS, and ignition data to reduce noise and ensure synchronization. The preprocessed data is then transmitted through LoRa as the primary communication channel, with GSM acting as a backup for essential alerts, and stored in a cloud-based backend. Anomaly detection is carried out in the cloud using machine learning models, including Matrix Profile (STUMPY) for time-series discord identification and Isolation Forest for feature-based outlier detection. Model validation and evaluation are performed using labeled and test datasets, calculating performance metrics such as precision, false positive rate (FPR), detection latency, and overall accuracy.

The system outputs include fuel anomaly flags, such as potential theft or leakage, and driver rest alerts based on cumulative distance or driving duration. Additionally, it provides real-time GPS tracking on a web dashboard, estimated fuel levels as a percentage of tank capacity, and quantitative performance metrics for operational evaluation. This end-to-end flow ensures accurate, real-time monitoring, proactive safety notifications, and actionable insights for fleet management.

5.10 Evaluation

To evaluate these models, the outputs are validated against manually labeled ground truth data. From this, the number of true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN) are counted. Key metrics are computed as follows:

Precision

$$\text{Precision} = \frac{TP}{TP + FP} \tag{5.1}$$

Precision reflects how many detected anomalies are truly anomalous. A high precision means fewer false alarms, which is important for maintaining trust in the alerting system.

False Positive Rate (FPR)

$$\text{FPR} = \frac{FP}{FP + TN} \tag{5.2}$$

FPR quantifies the likelihood of incorrectly classifying normal behavior as anomalous. Lower FPR values are desired to reduce noise in the system's alerts.

Accuracy Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{5.3}$$

Accuracy provides an overall view of model correctness but can be misleading in imbalanced datasets, which is why it's used alongside precision and FPR.

All these metrics use the following classification outcomes:

$$\begin{aligned}
\text{TP (True Positives)} &= \text{Number of correctly identified anomalies,} \\
\text{FP (False Positives)} &= \text{Number of normal events incorrectly flagged as anomalies,} \\
\text{TN (True Negatives)} &= \text{Number of correctly identified normal cases,} \\
\text{FN (False Negatives)} &= \text{Anomalies that were missed by the model.}
\end{aligned}$$

Detection Latency

$$L_{\text{det}} = T_{\text{detected}} - T_{\text{actual}} \tag{5.4}$$

where:

$$T_{\text{detected}} = \text{the time the anomaly truly occurred,}$$

$$T_{\text{actual}} = \text{the time it was identified by the system.}$$

The objective is achieved if the unsupervised system attains at least 90% precision (indicating that the majority of flagged events are correct), a false positive rate not exceeding 10%, and average detection latency under 10 seconds. For interpretability, confusion matrices and performance plots such as ROC and PR curves will be used. Detection events will also be overlaid on synchronized GPS location maps and fuel-level graphs for context-aware validation.

5.11 Summary

This chapter presented a comprehensive methodology for an AI-driven fuel monitoring and anomaly detection system. The approach integrates sensor-level calibration, edge preprocessing, IoT-based communication, cloud ML evaluation, and client-side visualization to achieve accurate, real-time, and actionable insights.

Chapter 6

RESULTS AND DISCUSSIONS

Show in this chapter proofs why your proposed solution works. However, presenting results ("It worked") without an appropriate explanation does not show thorough understanding. Aside from the data and results that you have obtained, and their explanation, the discussion includes why components of your proposed solution work did or did not work in accordance to what you described in the evaluation process, and how the proposed solution performed and faired. Interpret the results and the reasons why they were obtained. If your results are incorrect, apparent discrepancies from theory should be pointed out and explained. In essence, what do the results mean? Citing existing publication can help you compare your results and your explanations.

The next items below is not related to the description of this results and discussions chapter, but serves as an opener for the LaTeXportion of this template.

Here is an example of a citation for ISO 80000-2 standard [**?**]. Another one is [**?**] and [**?**].

In using this template, the user is expected to have a working knowledge of LaTeX. A good introduction is in [**?**]. Its latest version can be accessed at `http://www.ctan.org/tex-archive/info/lshort`. See the Appendix of `document_guide.pdf` for examples.

In aggregate form, Table 6.1 shows the outcomes and completions in applying the methodology of the Thesis Proposalper objective.

TABLE 6.1 SUMMARY OF RESULTS FOR ACHIEVING THE OBJECTIVES

Objectives	**Results**	**Locations**

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Objectives	**Results**	**Locations**
GO: To develop and evaluate an IoT-based system for trucks, in collaboration with the partner truck company Hino Parañaque Subic GS Auto Inc. (SGSA), that monitors real-time fuel levels, tracks routes using GPS, and detects unusual fuel consumption using machine learning, with the goal of improving driver safety, supporting rest scheduling, and facilitating proactive vehicle maintenance through automated alerts using fuel and distance metrics, based on established service guidelines.	1. First itemtext 2. Second itemtext 3. Last itemtext 4. First itemtext 5. Second itemtext	Sec. 5.9 on p. 111

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Objectives	**Results**	**Locations**
SO1: To measure fuel levels using digital float sensors integrated via the truck's ECU/diagnostic connector and GPS module, ensuring non-invasive installation and real-time accuracy within ±5%.	1. First itemtext 2. Second itemtext 3. Last itemtext 4. First itemtext 5. Second itemtext	Sec. 5.9 on p. 111
SO2: To transmit real-time fuel and GPS data through IoT-based communication to a cloud server, accessible through a centralized dashboard.	1. First itemtext 2. Second itemtext 3. Last itemtext 4. First itemtext 5. Second itemtext	Sec. 5.9 on p. 111
SO3: To capture and log GPS location data every 5 seconds via an embedded module, correlating route information with fuel and time data for behavior tracking.	1. First itemtext 2. Second itemtext 3. Last itemtext 4. First itemtext 5. Second itemtext	Sec. 5.9 on p. 111

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Objectives	Results	Locations
SO4: To design and evaluate a context-aware fuel usage deviation modeling approach using non-parametric unsupervised machine learning models, like Matrix Profile or Isolation Forest, achieving 90% precision and 10% false positives in identifying anomalous fuel consumption patterns within controlled environments.	1. First itemtext 2. Second itemtext 3. Last itemtext 4. First itemtext 5. Second itemtext	Sec. 5.9 on p. 111
SO5: To develop and integrate hardware-based rest and maintenance alert mechanisms (e.g., Start/Stop button, buzzer indicators) tied to the web dashboard using cumulative fuel and distance data.	1. First itemtext 2. Second itemtext 3. Last itemtext 4. First itemtext 5. Second itemtext	Sec. 5.9 on p. 111

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6.1 Summary

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Chapter 7

CONCLUSIONS, RECOMMENDATIONS, AND FUTURE DIRECTIVES

7.1 Concluding Remarks

In this Thesis Proposal, . . .

Put here the main points that should be known and learned about the work topic. Summarize or give the gist of the essential principles and inferences drawn from your results.

7.2 Contributions

The interrelated contributions and supplements that have been developed by the author(s) in this Thesis Proposal are listed as follows. Only those that are unique to the authors' work are included.

- the ;
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7.3 Recommendations

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7.4 Future Prospects

There are several prospects that may be extended for further studies. . . . So the suggested topics are listed in the following.

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Note that for ECE undergraduate theses, as per the directions of the thesis adviser, Recommendations and Future Directives will be removed for the hardbound copy but will be retained for database storage.

REFERENCES

[iec,] Iec 61131-2:2017. https://webstore.iec.ch/en/publication/31007. Accessed: 2025-11-27.

[iee,] Ieee p7003: Algorithmic bias considerations. https://standards.ieee.org/ieee/7003/11357/. Accessed: 2025-11-27.

[fcc,] Ieee xplore full-text pdf. https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10602499. Accessed: 2025-11-27.

[iso, a] Iso 26262-1:2011. https://www.iso.org/standard/43464.html. Accessed: 2025-11-27.

[iso, b] Iso 39001:2012. https://www.iso.org/standard/44958.html. Accessed: 2025-11-27.

[iso, c] Iso 45001:2018. https://www.iso.org/standard/63787.html. Accessed: 2025-11-27.

[iso, d] Iso 7637-2:2011. https://www.iso.org/standard/50925.html. Accessed: 2025-11-27.

[iso, e] Iso/iec 19510:2013. https://www.iso.org/standard/62652.html. Accessed: 2025-11-27.

[iso, f] Iso/iec 27018:2025. https://www.iso.org/standard/27018. Accessed: 2025-11-27.

[iso, g] Iso/iec/ieee 42010:2022. https://www.iso.org/standard/74393.html. Accessed: 2025-11-27.

[rfc,] Rfc 791: Internet protocol. https://datatracker.ietf.org/doc/html/rfc791. Accessed: 2025-11-27.

[iso, 2015] (2015). Iso 14001:2015 environmental management systems. https://www.iso.org/standard/60857.html. ISO.

[aut, 2019] (2019). Automatic fuel monitoring system. *International Journal of Recent Technology and Engineering*, 8(4S2):348–352.

[iso, 2020] (2020). Iso/ts 15143-3:2020: Earth-moving machinery and mobile machines — worksite data exchange — part 3: Telematics data. https://www.iso.org/standard/76394.html. ISO.

[cec, 2022] (2022). Ce marking – obtaining the certificate, eu requirements. https://europa.eu/youreurope/business/product-requirements/labels-markings/ce-marking/index_en.htm. Accessed: 2025-11-27.

[iso, 2022a] (2022a). Iso/iec 27001:2022 information security management systems. https://www.iso.org/standard/27001. ISO.

[iso, 2022b] (2022b). Iso/iec/ieee 29119-1:2022 software and systems engineering — software testing. https://www.iso.org/standard/81291.html. ISO.

[iso, 2023] (2023). Iso 50001 — energy management systems. https://www.iso.org/iso-50001-energy-management.html. ISO.

[iso, 2024] (2024). Iso/pas 8800:2024 road vehicles — safety and artificial intelligence. https://www.iso.org/standard/83303.html. ISO.

[Ahmed et al., 2017] Ahmed, A. A. I., Mohammed, S. A. E., and Satte, M. A. M. H. (2017). Fuel management system. In *2017 International Conference on Communication, Control, Computing and Electronics Engineering (ICCCCEE)*, pages 1–7.

[Alquhali et al., 2019] Alquhali, A. H., Roslee, M., Alias, M. Y., and Mohamed, K. S. (2019). IoT based real-time vehicle tracking system. In *2019 IEEE Conference on Sustainable Utilization and Development in Engineering and Technologies (CSUDET)*, pages 265–270.

[Alsanad et al., 2020] Alsanad, H. R., Ucan, O. N., Ilyas, M., Khan, A. U. R., and Bayat, O. (2020). Real-time fuel truck detection algorithm based on deep convolutional neural network. *IEEE Access*, 8:118808–118817.

[Aquize et al., 2017] Aquize, V. G., Emery, E., and de Lima Neto, F. B. (2017). Self-organizing maps for anomaly detection in fuel consumption. case study: Illegal fuel storage in bolivia. In *2017 IEEE Latin American Conference on Computational Intelligence (LA-CCI)*, pages 1–6.

[Arlot and Celisse, 2010] Arlot, S. and Celisse, A. (2010). A survey of cross-validation procedures for model selection. *Statistics Surveys*, 4:40–79.

[Asian Development Bank, 2016] Asian Development Bank (2016). Fuel smuggling: 12 things to know. https://www.adb.org/news/features/fuel-smuggling-12-things-know. Accessed: 2026-06-21.

[Barbado and Corcho, 2022] Barbado, A. and Corcho, Ó. (2022). Interpretable machine learning models for predicting and explaining vehicle fuel consumption anomalies. *Engineering Applications of Artificial Intelligence*, 115:105222.

[Beteto et al., 2022] Beteto, A. et al. (2022). Anomaly and cyber fraud detection in pipelines and supply chains for liquid fuels. *Environment Systems and Decisions*, 42(2):306–324.

[Chandola et al., 2009] Chandola, V., Banerjee, A., and Kumar, V. (2009). Anomaly detection. *ACM Computing Surveys*, 41(3):1–58.

[Chengdu Ebyte Electronic Technology Co., Ltd., 2024] Chengdu Ebyte Electronic Technology Co., Ltd. (2024). Lora module range: Key factors affecting lora communication. https://www.cdebyte.com/news/622. Accessed: 2026-06-19.

[Cherdo et al., 2023] Cherdo, Y., Miramond, B., Pegatoquet, A., and Vallauri, A. (2023). Unsupervised anomaly detection for cars can sensors time series using small recurrent and convolutional neural networks. *Sensors*, 23(11):5013–5013.

[China Academy of Information and Communications Technology, 2025] China Academy of Information and Communications Technology (2025). Certification updates of NTC certification of philippines. https://gma.caict.ac.cn/en/update/31. Accessed: 2026-06-24.

[Crisgar et al., 2021] Crisgar, P. V. et al. (2021). GPS-based vehicle tracking and theft detection systems using google cloud IoT core & firebase. In *2021 International Symposium on Electronics and Smart Devices (ISESD)*, pages 1–6.

[Davis and Goadrich, 2006] Davis, J. and Goadrich, M. (2006). The relationship between precision-recall and ROC curves. In *Proceedings of the 23rd International Conference on Machine Learning*, pages 233–240.

[Devalal and Karthikeyan, 2018] Devalal, S. and Karthikeyan, A. (2018). Lora technology - an overview. In *2018 Second International Conference on Electronics, Communication and Aerospace Technology (ICECA)*, pages 284–290, Coimbatore, India.

[Dhivyasri and Mariappan, 2015] Dhivyasri, G. and Mariappan, R. (2015). Advanced fuel level tracking system using telematic unit. In *2015 IEEE International Conference on Electrical, Computer and Communication Technologies (ICECCT)*.

[Emmott et al., 2015] Emmott, A. F., Das, S., Dietterich, T., Fern, A., and Wong, W.-K. (2015). A systematic construction of anomaly detection benchmarks from real data. In *Proceedings of the ACM SIGKDD Workshop on Outlier Detection and Description*, pages 16–21.

[Farzana, 2020] Farzana, F. e. a. (2020). A real-time motion based fuel monitoring technique for vehicle tracking systems. In *2020 Emerging Technology in Computing, Communication and Electronics (ETCCE)*, pages 1–6.

[Fawcett, 2006] Fawcett, T. (2006). An introduction to ROC analysis. *Pattern Recognition Letters*, 27(8):861–874.

[Fleet Complete, 2026] Fleet Complete (2026). Fleet management and fuel reporting. https://www.fleetcomplete.com/. Accessed: 2026-06-21.

[Fleetio, 2026] Fleetio (2026). Fuel management system. https://www.fleetio.com/solutions/fuel-management-software. Accessed: 2026-06-08.

[Fortela et al., 2023] Fortela, D. L. B. et al. (2023). Unsupervised machine learning to detect impending anomalies in testing of fuel economy and emissions of light-duty vehicles. *Clean Technologies*, 5(1):418–435.

[G et al., 2022] G, P. et al. (2022). Real time automatic vehicle monitoring system using IoT. In *2022 8th International Conference on Smart Structures and Systems (ICSSS)*, pages 1–6.

[G, 2023] G, Y. e. a. (2023). Fuel monitoring in bike vehicles based on iot. In *2023 Intelligent Computing and Control for Engineering and Business Systems (ICCEBS)*, pages 1–5.

[Garcia Vega, 2022] Garcia Vega, P. (2022). Development of a method for anomaly detection in time series applied to vehicle monitoring. Master's thesis, Polytechnique Montréal.

[Geotab, 2024] Geotab (2024). Fleet fuel management systems. https://www.geotab.com/fleet-management-solutions/fleet-fuel-management/. Accessed: 2026-06-09.

[GSMA, 2026] GSMA (2026). Network coverage maps. https://www.gsma.com/coverage/#433. Accessed: 2026-06-19.

[Habib and Umar, 2015] Habib, K. and Umar, A. I. (2015). Anomalies calculation and detection in fuel expense through data mining. *Research Journal of Information Technology*, 6(4):44–50.

[Hirth et al., 2025] Hirth, M., Meier, D., Gehring, O., Jazdi, N., and Kasneci, E. (2025). Unsupervised deep learning for anomaly detection in automotive trucks: A survey. *ACM Computing Surveys*, 58(5).

[Husak,] Husak, V. e. a. Intelligent real-time vehicle tracking information system. n.d.

[IEEE Standards Association,] IEEE Standards Association. Ieee standards association. https://standards.ieee.org/ieee/1451.0/11001/. Accessed: 2025-11-27.

[International Organization for Standardization, 2015] International Organization for Standardization (2015). ISO 15031-5:2015 road vehicles – communication between vehicle and external equipment for emissions-related diagnostics – part 5: Emissions-related diagnostic services. https://www.iso.org/standard/66368.html.

[International Organization for Standardization, 2021] International Organization for Standardization (2021). ISO 15765-4:2021 road vehicles – diagnostic communication over controller area network (DoCAN) – part 4: Requirements for emissions-related systems. https://www.iso.org/standard/78384.html.

[International Organization for Standardization, 2024] International Organization for Standardization (2024). ISO 11898-1:2024 road vehicles – controller area network (CAN) – part 1: Data link layer and physical coding sublayer. https://www.iso.org/standard/86384.html. Accessed: 2026-06-23.

[Japan International Cooperation Agency, 2022] Japan International Cooperation Agency (2022). The project for comprehensive traffic management plan for metro manila final report main text. Technical report, Japan International Cooperation Agency.

[Japan International Cooperation Agency, 2025] Japan International Cooperation Agency (2025). Jica and mmda commence a project on intelligent transportation system in line with the comprehensive traffic management plan for metro manila. https://www.jica.go.jp/english/overseas/philippine/information/press/2024/rdsigningctmp02212025.html. Accessed: 2026-06-21.

[Jose, 2015] Jose, D. e. a. (2015). Intelligent vehicle monitoring using global positioning system and cloud computing. *Procedia Computer Science*, 50:440–446.

[Joshi et al., 2024] Joshi, P. et al. (2024). Intelligent IoT-enabled real-time monitoring system for logistics management. In *2024 International Conference on Computing, Sciences and Communications (ICCSC)*, pages 1–5.

[Kanchi et al., 2013] Kanchi, S., Sandilya, S., Bhosale, D., Pitkar, A., and Gondhalekar, M. (2013). Overview of lte-a technology. In *2013 IEEE Global High Tech Congress on Electronics*, pages 195–200, Shenzhen, China.

[Komal Shoukat Ali et al., 2018] Komal Shoukat Ali, K. et al. (2018). Automatic fuel tank monitoring, tracking & theft detection system. In *MATEC Web of Conferences*, volume 184.

[Krishnasamy, 2019] Krishnasamy, R. (2019). Automatic fuel monitoring system. *International Journal of Recent Technology and Engineering (IJRTE)*, 8(4S2):348–352. Accessed: 2025-11-18.

[Lavin and Ahmad, 2015] Lavin, A. and Ahmad, S. (2015). Evaluating real-time anomaly detection algorithms – the Numenta anomaly benchmark. In *2015 IEEE 14th International Conference on Machine Learning and Applications*, pages 38–44.

[Lee et al., 2007] Lee, J. S., Su, Y. W., and Shen, C. C. (2007). A comparative study of wireless protocols: Bluetooth, uwb, zigbee, and wi-fi. In *Proceedings of the 33rd Annual Conference of the IEEE Industrial Electronics Society*, China.

[Liu et al., 2008] Liu, F. T., Ting, K. M., and Zhou, Z.-H. (2008). Isolation forest. In *2008 Eighth IEEE International Conference on Data Mining*, pages 413–422, Pisa, Italy.

[Llanto, 2017] Llanto, G. (2017). Cargo truck ban: Bad timing, faulty analysis, policy failure. Technical report, Philippine Institute for Development Studies.

[Lu et al., 2022] Lu, Y., Wu, R., Mueen, A., Zuluaga, M. A., and Keogh, E. (2022). Matrix profile xxiv: Scaling time series anomaly detection to trillions of datapoints and ultra-fast arriving data streams. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*.

[M, 2021] M, S. e. a. (2021). Digital fuel monitoring system for automobiles. In *2021 Second International Conference on Electronics and Sustainable Communication Systems (ICESC)*, pages 6–10.

[Majeed, 2015] Majeed, A. (2015). Comparative studies of 3g, 4g & 5g mobile network & data offloading method a survey. *International Journal of Research in Information Technology*.

[Manalu et al., 2023] Manalu, I. P., Sigiro, M. M. T., Wowiling, G. I., and Simatupang, F. (2023). Lora performance analysis with study case a moving object. *International Journal of Engineering Research in Computer Science and Engineering*, 10(12):11–16.

[Mathi et al., 2024] Mathi, S. C. et al. (2024). Development of an intelligent fuel monitoring and theft detection system equipped with GPS & GSM integration, state of charge monitoring and realtime alarm generation. In *2024 International Conference on Innovative Computing and Communication Technologies (ICICT)*, pages 2095–2102.

[Maurya,] Maurya, K. e. a. Real time vehicle tracking system using gsm and gps technology- an anti-theft tracking system. n.d.

[Meitrack,] Meitrack. Lora-based fleet tracking solution. https://www.meitrack.com/lora-based-fleet-tracking-solution/. Accessed: 2026-06-19.

[Miyim and Wakili, 2019] Miyim, A. and Wakili, A. (2019). Performance evaluation of lte networks. In *2019 15th International Conference on Electronics, Computer and Computation (ICECCO)*.

[Mukesh, 2010] Mukesh, P. R. (2010). Real time web based vehicle tracking using gps.

[Mumcuoglu et al., 2023] Mumcuoglu, M. E. et al. (2023). Fuel consumption classification for heavy-duty vehicles: A novel approach to identifying driver behavior and system anomalies. In *2023 AEIT International Conference on Electrical and Electronic Technologies for Automotive (AEIT AUTOMOTIVE)*, pages 1–6.

[Nada, 2017] Nada, Y. A. (2017). Design and implementation for trucks tracking system using gps based on semantic web. *International Journal of Trend in Scientific Research and Development*, 1(4).

[NASA EarthData, 2025] NASA EarthData (2025). Global navigation satellite system. https://www.earthdata.nasa.gov/data/space-geodesy-techniques/gnss. Accessed: 2025-11-27.

[Navixy, 2026] Navixy (2026). Sensors setup and configuration. https://www.navixy.com/docs/expert-center/vehicle-telematics-technology/fuel-management/fuel-control-in-navixy/sensors-setup-and-configuration. Accessed: 2026-06-08.

[Obikoya, 2014] Obikoya, G. D. (2014). Design, construction, and implementation of a remote fuel-level monitoring system. *EURASIP Journal on Wireless Communications and Networking*, 2014(1):76.

[Opensignal,] Opensignal. Se asia - coverage. https://www.tutela.com/state-of-mobile-networks-southeast-asia/coverage. Accessed: 2026-06-19.

[Orange Connected Objects & Partnerships, 2016] Orange Connected Objects & Partnerships (2016). *LoRa Device Developer Guide*.

[Patalinghug et al., 2015a] Patalinghug, E. et al. (2015a). Port congestion and underutilization in the greater capital region: Unpacking the issues. Technical report, Philippine Institute for Development Studies.

[Patalinghug et al., 2015b] Patalinghug, E. et al. (2015b). A system-wide study of the logistics industry in the greater capital region. Technical report, Philippine Institute for Development Studies.

[Patil and Patil, 2024] Patil, A. and Patil, S. (2024). Iot enabled vehicle monitoring and theft prevention system (i-vmtps). In *2024 2nd International Conference on Advancements and Key Challenges in Green Energy and Computing (AKGEC)*, pages 1–7.

[Powers, 2020] Powers, D. M. W. (2020). Evaluation: From precision, recall and F-measure to ROC, informedness, markedness and correlation. *arXiv*. Accessed: 2026-06-24.

[Prabhu et al., 2022] Prabhu, S. B., Deepa, K., and Nithya, M. (2022). IoT enabled fuel level monitoring and automatic fuel theft detection system. In *2022 13th International Conference on Computing Communication and Networking Technologies (ICCCNT)*.

[Prasanna et al., 2023] Prasanna, J. L., Santhosh, C., Kumar, M. R., and Gayathri, K. (2023). Iot based automatic fuel level monitoring system for automobiles. In *IET Conference Proceedings*, volume 2023, pages 56–61.

[Robert Bosch GmbH, 1991] Robert Bosch GmbH (1991). *CAN Specification Version 2.0*. Robert Bosch GmbH. Accessed: 2026-06-23.

[RS et al., 2024] RS, S., Narayanan CS, S., G, R., S, P., Alagarsamy, M., and William, P. (2024). Fuel quantity and quality detection in automobiles using IoT. In *2024 4th International Conference on Innovative Practices in Technology and Management (ICIPTM)*, pages 1–6.

[S et al., 2024] S, S., Sowmya, J. M., Siva, M. V., Rishitha, N., Sagar, M. K. G. S., and Ganesh, M. (2024). Iot enabled smart fuel station management system. In *2024 2nd International Conference on Networking and Communications (ICNWC)*, pages 1–5.

[SAE International, 2016] SAE International (2016). J1962_201607: Diagnostic connector. https://www.sae.org/standards/content/j1962_201607/. Accessed: 2026-06-23.

[SAE International, 2017] SAE International (2017). J1979_201702: E/E diagnostic test modes. https://saemobilus.sae.org/standards/j1979_201702-e-e-diagnostic-test-modes. Accessed: 2026-06-23.

[SAE International, 2023] SAE International (2023). SAE J1939: Serial control and communications heavy duty vehicle network. https://www.sae.org/standards/j1939_202306-serial-control-communications-heavy-duty-vehicle-network-top-level-document.

[SAE International, 2024] SAE International (2024). SAE J1939DA: Digital annex of serial control and communications heavy duty vehicle network.

[Saito and Rehmsmeier, 2015] Saito, T. and Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, 10(3):e0118432.

[Samainstage, 2025] Samainstage (2025). The evolution of wi-fi technology and standards. https://standards.ieee.org/beyond-standards/the-evolution-of-wi-fi-technology-and-standards/. Accessed: 2025-11-27.

[Samsara, 2026] Samsara (2026). Fleet fuel management. https://www.samsara.com/products/telematics/fleet-fuel-management-system. Accessed: 2026-06-09.

[scikit-learn developers,] scikit-learn developers. Leaveonegroupout. https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.LeaveOneGroupOut.html. Accessed: 2026-06-25.

[Semenov et al., 2024] Semenov, I., Świderski, A., Borucka, A., and Guzanek, P. (2024). Concept of early prediction and identification of truck vehicle failures supported by in-vehicle telematics platform based on abnormality detection algorithm. *Applied Sciences (Switzerland)*, 14(16).

[Tatababu et al., 2024] Tatababu, D., Ranga, M. R., Satyannarayana, D., Jyothi, Y., and Devi, A. R. (2024). Smart fuel monitoring system for diesel generator. In *2024 International Conference on Computational Intelligence for Green and Sustainable Technologies (ICCIGST)*, pages 1–6.

[Thajudheen et al., 2023] Thajudheen, S., Shanmugarathinum, G., and Jesudoss, A. G. (2023). Vehicular data retrieval system. In *2023 International Conference on New Frontiers in Communication, Automation, Management and Security (ICCAMS)*.

[The International Engineering Consortium,] The International Engineering Consortium. *Global System for Mobile Communication (GSM)*. Web ProForum Tutorials. Tutorial.

[Turan et al., 2024] Turan, B. B., Gene, E., Akcig, I. N., Goztepe, N., Mumeuoglu, M. E., and Unel, M. (2024). Detecting high fuel consumption in hdvs with ensemble of anomaly detection models.

In *2024 IEEE 22nd International Conference on Industrial Informatics (INDIN)*, pages 1–6.

[Uddin et al., 2017] Uddin, M. S., Ahmed, M. M., Alam, J. B., and Islam, M. (2017). Smart anti-theft vehicle tracking system for bangladesh based on internet of things. In *2017 4th International Conference on Advances in Electrical Engineering (ICAEE)*, pages 624–628.

[Usama et al., 2019] Usama, M., Qadir, J., Raza, A., Arif, H., Yau, K.-L. A., Elkhatib, Y., Hussain, A., and Al-Fuqaha, A. (2019). Unsupervised machine learning for networking: Techniques, applications and research challenges. *IEEE Access*, 7:65579–65615.

[Vigneshwaran et al., 2020] Vigneshwaran, S., Nithya, B., Raghul, K., Nivas, B., and Kishore, V. M. (2020). Design of bus tracking and fuel monitoring system. In *2020 6th International Conference on Advanced Computing and Communication Systems (ICACCS)*, pages 348–351.

[Weiwei et al., 2024] Weiwei, Q., Longbin, L., Xingyu, W., Yulong, L., and Weiqian, D. (2024). Research on diesel consumption prediction and early warning analysis of dump trucks in smart mines. In *2024 IEEE 22nd International Conference on Industrial Informatics (INDIN)*, pages 1–6.

[Wialon, 2026] Wialon (2026). Fuel control. https://wialon.com/en/fuel-control. Accessed: 2026-06-21.

[Wilson, 1927] Wilson, E. B. (1927). Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212.

[Yeh et al., 2016] Yeh, C.-C. M. et al. (2016). Matrix profile i: All pairs similarity joins for time series: A unifying view that includes motifs, discords and shapelets. In *2016 IEEE 16th International Conference on Data Mining (ICDM)*, pages 1317–1322, Barcelona, Spain.

[Yeh et al., 2024] Yeh, C.-C. M. et al. (2024). Matrix profile for anomaly detection on multidimensional time series. In *2024 IEEE International Conference on Data Mining (ICDM)*, pages 911–916, Abu Dhabi, United Arab Emirates.